THE EXACT ROSENTHAL BOUND FOR INDEPENDENT ZERO-MEAN SUMMANDS WHEN $3 \le p \le 4$, AND IN THE TWO EXTREME REGIMES WHEN $4 < p < 5$

ABSTRACT. For $p > 2$, $A, B > 0$, let $\mathcal{E}_{p;A,B}$ be the supremum of $\mathbb{E}|X_1 + \cdots + X_n|^p$ over all n and all independent zero-mean random variables with $\sum \mathbb{E}X_i^2 = B$ and $\sum \mathbb{E}|X_i|^p = A$. Pinelis (Ann. Probab. 43 (2015)) computed $\mathcal{E}_{p;A,B}$ for $p \in (2,3] \cup [5,\infty)$ and stated that the cases $p \in (3,4)$ and $p \in (4,5)$ remain open. We prove that $\mathcal{E}_{p;A,B} = A + \mathbb{E}|Z|^p B^{p/2}$ for all $p \in [3,4]$, $Z \sim N(0,1)$. In other words,

$$\mathbb{E}\Big|\sum_{i=1}^n X_i\Big|^p \le \sum_{i=1}^n \mathbb{E}|X_i|^p + \mathbb{E}|Z|^p \Big(\sum_{i=1}^n \mathbb{E}X_i^2\Big)^{p/2}, \qquad 3 \le p \le 4,$$

with the sharp constants; in the max form this confirms a hypothesis of Ibragimov and Sharakhmetov (2002) for $3 < t < 4$. The proof combines Utev's Poissonization with an elementary Fourier identity whose integrand is pointwise nonpositive; a second proof reduces the zero-mean case to the classical symmetric one via a symmetrization inequality for Lévy measures. Consequences include sharp versions for Hilbert-space-valued and L^p-valued summands and for a class of general moment functions. For $3 \le p \le 4$ the same argument gives the sharp bound $A + \mathbb{E}|X + B^{1/2}Z|^p$ in the presence of an extra independent summand X whose characteristic function is nonnegative. On the other hand, the extension with a general extra summand, suggested by Pinelis (his Theorem 1.5 for $p \in (3,4)$), is false: it fails for $X \equiv 1$ and for some bounded zero-mean X. For $4 < p < 5$, with Π_λ a Poisson variable with mean λ and $c^p\lambda = A$, $c^2\lambda = B$, we prove $c^p\mathbb{E}|\Pi_\lambda - \lambda|^p \le \mathcal{E}_{p;A,B} \le c^p\mathbb{E}|\Pi_\lambda - \lambda|^p + C_p B^{p/2}$, and we prove that Pinelis' conjectured Poisson value is exact, $\mathcal{E}_{p;A,B} = c^p\mathbb{E}|\Pi_\lambda - \lambda|^p$, whenever $B^{p/2}/A$ is sufficiently small (the threshold produced by the proof is very small), and likewise whenever $B^{p/2}/A$ is sufficiently large (again with a large threshold); so the conjecture holds near both ends of the parameter range. Quantitatively, for $\lambda \ge 1$ the supremum exceeds the Poisson value $B^{p/2}(m_p + K_p\lambda^{-1} + O(\lambda^{-2}))$, $K_p = \binom{p}{3}\mathbb{E}|Z|^p/12$, by at most $O(B^{p/2}\lambda^{-2})$. The Gaussian-regime results use a decomposition whose leading part is a linear functional of the Lévy measure maximized at the Poisson law, by a concavity (two-moment-problem) argument. We also reduce the full conjecture to a "skewness inequality" for compound Poisson laws, supported by numerics, and exhibit the non-convexity phenomenon that blocks second-variation arguments there.

1. Statement of results

Rosenthal's inequality [9] bounds $\mathbb{E}|\sum X_i|^p$ by a constant times $\max\{\sum \mathbb{E}|X_i|^p, (\sum \mathbb{E}X_i^2)^{p/2}\}$; here we are concerned with exact (sharp) forms. Throughout, $Z \sim N(0,1)$, $m_p := \mathbb{E}|Z|^p = 2^{p/2}\Gamma(\frac{p+1}{2})/\sqrt{\pi}$, and $\binom{p}{k} := p(p-1)\cdots(p-k+1)/k!$ for real p. For $p > 2$ and $A, B > 0$ let $\mathscr{X}_{p;A,B}$ be the set of all finite sequences $\mathbf{X} = (X_1, \ldots, X_n)$ of independent random variables with $\mathbb{E}X_i = 0$, $\sum_i \mathbb{E}X_i^2 = B$, $\sum_i \mathbb{E}|X_i|^p = A$, and put $S_{\mathbf{X}} = X_1 + \cdots + X_n$,

$$\mathcal{E}_{p;A,B} := \sup\{\mathbb{E}|S_{\mathbf{X}}|^p : \ \mathbf{X} \in \mathscr{X}_{p;A,B}\}.$$

Theorem A. *Let $3 \le p \le 4$. For every $n \ge 1$ and all independent zero-mean $X_1, \ldots, X_n$ with $\mathbb{E}|X_i|^p < \infty$,*

$$\mathbb{E}|X_1 + \cdots + X_n|^p \le \sum_{i=1}^n \mathbb{E}|X_i|^p + m_p\Big(\sum_{i=1}^n \mathbb{E}X_i^2\Big)^{p/2}. \tag{1}$$

Moreover $\mathcal{E}_{p;A,B} = A + m_p B^{p/2}$ for all $A, B > 0$.

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Together with Pinelis' Theorem 1.5 [7] (the range $2 < p \le 3$, proved there via Tyurin's zero-bias bound [10]) this gives $\mathcal{E}_{p;A,B} = A + m_p B^{p/2}$ on the whole range $2 < p \le 4$. Consequently, the best constant Δ_p in $\|S_{\mathbf{X}}\|_p \le \Delta_p \max\{(\sum \mathbb{E}X_i^2)^{1/2}, (\sum \mathbb{E}|X_i|^p)^{1/p}\}$ for independent *zero-mean* summands is $\Delta_p = (1+m_p)^{1/p}$ for $2 < p \le 4$. For $3 < p < 4$ this confirms the hypothesis stated at the end of [6]: for $2 < t < 4$, the exact constant $C(t)$ in $\mathbb{E}|S|^t \le C(t) \max\{\sum \mathbb{E}|X_i|^t, (\sum \mathbb{E}X_i^2)^{t/2}\}$ for zero-mean summands coincides with the constant $1 + m_t$ for symmetric summands. The range $2 < t \le 3$ follows from [7, Thm. 1.5]. This is the same value as for symmetric summands ([4, Thm. 6.1, Cor. 6.2], [5]). Indeed, by homogeneity $\mathcal{E}_{p;A,B} \le (1+m_p)\max\{A, B^{p/2}\}$ with equality at $A = B^{p/2}$.

Corollary B (Hilbert-space-valued summands)**.** *Let $2 < p \le 4$, let $\mathbb{H}$ be a separable real Hilbert space, and let $X_1, \dots, X_n$ be independent $\mathbb{H}$-valued random vectors with $\mathbb{E}X_i = 0$ and $\mathbb{E}\|X_i\|^p < \infty$. Let Γ be a centered Gaussian vector in $\mathbb{H}$ whose covariance operator is the sum of the covariance operators of the X_i. Then*

$$\mathbb{E}\Big\| \sum_i X_i \Big\|^p \le \sum_i \mathbb{E}\|X_i\|^p + \mathbb{E}\|\Gamma\|^p \le \sum_i \mathbb{E}\|X_i\|^p + m_p \Big(\sum_i \mathbb{E}\|X_i\|^2 \Big)^{p/2}.$$

In particular the optimal constant in the max-form Rosenthal inequality for independent zero-mean summands in a Hilbert space is $(1+m_p)^{1/p}$ for $2 < p \le 4$.

Corollary C (L^p-valued summands)**.** *Let $2 < p \le 4$, let $(\Omega', \mathcal{F}', \nu)$ be a σ-finite measure space with $L^p(\nu)$ separable, and let $X_1, \dots, X_n$ be independent random elements of $L^p(\nu)$ (Borel measurable maps into $L^p(\nu)$), represented by jointly measurable functions $X_i(\omega, \omega')$, with $\mathbb{E}X_i(\cdot, \omega') = 0$ for ν-a.e. ω' and $\mathbb{E}\|X_i\|_p^p < \infty$. Then*

$$\mathbb{E}\Big\| \sum_i X_i \Big\|_{L^p(\nu)}^p \le \sum_i \mathbb{E}\|X_i\|_{L^p(\nu)}^p + m_p \Big\| \Big(\sum_i \mathbb{E}|X_i|^2 \Big)^{1/2} \Big\|_{L^p(\nu)}^p,$$

and the constants 1 and m_p are optimal.

Proof. Independence of sections. For bounded continuous $\varphi_1, \dots, \varphi_n$ on $\mathbb{R}$ and $E \in \mathcal{F}'$ with $\nu(E) < \infty$, the map $\Psi(x_1, \dots, x_n) := \int_E \prod_i \varphi_i(x_i(\omega'))\nu(d\omega')$ is continuous on $L^p(\nu)^n$ (from every sequence converging in L^p extract an a.e. convergent subsequence and use dominated convergence). By independence, $\mathbb{E}\Psi(X_1, \dots, X_n)$ equals the integral of Ψ against the product of the laws of the X_i; by Fubini (joint measurability) this reads $\int_E \mathbb{E}\prod_i \varphi_i(X_i(\cdot, \omega'))\,\nu(d\omega') = \int_E \prod_i \mathbb{E}\varphi_i(X_i(\cdot, \omega'))\,\nu(d\omega')$. As E is arbitrary and ν is σ-finite, the integrands agree ν-a.e. Applying this to the countably many tuples with $\varphi_i \in \{\cos(q\,\cdot), \sin(q\,\cdot) : q \in \mathbb{Q}\}$ shows that, for ν-a.e. ω', the joint characteristic function of $(X_i(\cdot, \omega'))_i$ factorizes at rational points, hence everywhere by continuity; so for ν-a.e. ω' the real random variables $X_i(\cdot, \omega')$ are independent and zero-mean.

Conclusion. Apply Theorem A (for $3 \le p \le 4$) or [7, Thm. 1.5] (for $2 < p \le 3$) to them, integrate in ω' against ν, and use Tonelli's theorem. The hypotheses of the scalar results hold for ν-a.e. ω', because $\int \mathbb{E}|X_i(\cdot, \omega')|^p\,\nu(d\omega') = \mathbb{E}\|X_i\|_p^p < \infty$. Optimality follows by taking ν a point mass. □

Theorem D. *Let $3 \le p \le 4$ and let X be a random variable with $\mathbb{E}|X|^p < \infty$ whose characteristic function is nonnegative, $\mathbb{E}e^{itX} \ge 0$ for all $t \in \mathbb{R}$ (e.g. $X = Y - Y'$ with Y' an independent copy of Y, or X centered Gaussian). If $X, X_1, \dots, X_n$ are independent and the X_i are zero-mean with $\sum \mathbb{E}X_i^2 = B$, $\sum \mathbb{E}|X_i|^p = A$, then*

$$\mathbb{E}|X + X_1 + \cdots + X_n|^p \le A + \mathbb{E}|X + B^{1/2}Z|^p,$$

and the right-hand side is the supremum over $\mathscr{X}_{p;A,B}$ (with Z independent of X).

Pinelis [7, Sect. 1, after Cor. 1.7] wrote that "certain considerations suggest that Theorem 1.5 should hold for $p \in (3, 4)$ as well". His Theorem 1.5 states, for $2 < p \le 3$ and every X with

$\mathbb{E}|X|^p < \infty$ (the condition $\mathbb{E}X = 0$ is not needed), that $\sup \mathbb{E}|X + S_{\mathbf{X}}|^p = A + \mathbb{E}|X + B^{1/2}Z|^p$, the supremum being over $\mathbf{X} \in \mathscr{X}_{p;A,B}$ independent of X (equivalently over the classes with $\le A$, $\le B$). Theorem A is the case $X = 0$. In the generality of an arbitrary extra summand the suggestion is false:

Theorem E. *Let $3 < p < 4$.*

(a) *For every $x_0 \neq 0$ there is a zero-mean two-point random variable X_1 such that, with $A = \mathbb{E}|X_1|^p$, $B = \mathbb{E}X_1^2$,*
$$\mathbb{E}|x_0 + X_1|^p > A + \mathbb{E}|x_0 + B^{1/2}Z|^p.$$
(b) *More generally, let X be a bounded random variable with $\mathbb{P}(|X| < \delta) = 0$ for some $\delta > 0$ and $\mathbb{E}[|X|^{p-3} \operatorname{sign} X] \neq 0$ (for instance X zero-mean two-point, $\mathbb{P}(X = -1) = \frac{3}{4}$, $\mathbb{P}(X = 3) = \frac{1}{4}$). Then there is a zero-mean two-point X_1, independent of X, with $\mathbb{E}|X + X_1|^p > \mathbb{E}|X_1|^p + \mathbb{E}|X + (\mathbb{E}X_1^2)^{1/2}Z|^p$.*

In particular the conclusion of [7, Thm. 1.5] *fails for every $p \in (3, 4)$, already with one summand.*

The proof of Theorem A uses only two properties of $|x|^p$, namely the Fourier representation of Lemma 2.1 and convexity of the second derivative. This answers, for a class of moment functions, the interest in general moment functions expressed in [7, Remark 1.4]:

Theorem F. *Let $\kappa \ge 0$, let μ be a nonnegative Borel measure on $(0, \infty)$ with $\int (t^2 \wedge t^4)\, \mu(dt) < \infty$, and put*
$$f(x) := \kappa x^4 + \int_{(0,\infty)} \Big(\cos tx - 1 + \frac{t^2x^2}{2} \Big) \mu(dt), \qquad x \in \mathbb{R}.$$
Assume that f'' is convex on $\mathbb{R}$. Then for all independent zero-mean $X_1, \dots, X_n$ with $\mathbb{E}X_i^4 < \infty$ and $B = \sum \mathbb{E}X_i^2$,
$$\mathbb{E}f(X_1 + \cdots + X_n) \le \sum_{i=1}^n \mathbb{E}f(X_i) + \mathbb{E}f(B^{1/2}Z).$$

Equivalently, f is even with $f(0) = f'(0) = 0$ and f'' is a convex, continuous, negative-definite function vanishing at 0: by the Lévy–Khintchine formula such f'' has the form $\sigma^2 x^2/2 + \int (1 - \cos tx)\, \nu(dt)$ with $\int (1 \wedge t^2) d\nu < \infty$, and then $\kappa = \sigma^2/24$, $\mu = t^{-2}\nu$, $\int (t^2 \wedge t^4) d\mu = \int (1 \wedge t^2) d\nu$. Examples: $|x|^p$ for $3 \le p \le 4$; finite mixtures $\int_{[3,4]} |x|^p \pi(dp)$ (for $p < 4$, μ has density $c_p^{-1} t^{-p-1}$ and $\int (t^2 \wedge t^4) c_p^{-1} t^{-p-1} dt = c_p^{-1}(\frac{1}{4-p} + \frac{1}{p-2})$ stays bounded as $p \uparrow 4$ because $c_p^{-1} \sim 24(4-p)$; the atom at $p = 4$ is the κx^4 term); and the function with $f''(x) = \sqrt{1 + x^2} - 1 = \int_0^\infty (1 - \cos tx) \frac{2K_1(t)}{\pi t} dt$ (the normal-inverse-Gaussian exponent; K_1 a modified Bessel function), which is convex and gives $f(x) \sim |x|^3/6$.

Finally, the identity behind Theorem A reverses beyond $p = 4$:

Proposition G. *Let $4 < p < 6$ and let X_G be a compound Poisson random variable with finite Lévy measure G on $\mathbb{R} \setminus \{0\}$, $\int x\, G(dx) = 0$, $\int x^2 G(dx) = B$, $\int |x|^p G(dx) = A < \infty$. Then $\mathbb{E}|X_G|^p \ge A + m_p B^{p/2}$.*

For $4 < p < 5$, where Pinelis' Poisson conjecture is open, we obtain a uniform two-sided bound that is sharp to second order in the rare-jump regime:

Theorem H. *Let $4 < p < 5$, $A, B > 0$, and let $c, \lambda > 0$ be given by $c^p\lambda = A$, $c^2\lambda = B$ (so $\lambda = (B^{p/2}/A)^{2/(p-2)}$ and $c^p\lambda^{p/2} = B^{p/2}$). Put $q_p := 2^{p-1} - p - 1$, $c_p' := -\Gamma(-p) \cos \frac{\pi p}{2} > 0$ and $\eta_p := \frac{12^{(6-p)/2}}{48(6-p)} + \frac{12^{(4-p)/2}}{4(p-4)}$. Then*
$$\big| c^p \mathbb{E}|\Pi_\lambda - \lambda|^p - A - q_p A\lambda \big| \le \frac{\eta_p}{c_p'} B^{p/2}, \qquad c^p \mathbb{E}|\Pi_\lambda - \lambda|^p \le \mathcal{E}_{p;A,B} \le A + q_p A\lambda + \frac{\eta_p}{c_p'} B^{p/2}.$$

In particular $\mathcal{E}_{p;A,B} \le c^p\mathbb{E}|\Pi_\lambda-\lambda|^p + C_pB^{p/2}$ *with* $C_p := 2\eta_p/c'_p$ *(*$C_{4.2}\approx 13.9$*,* $C_{4.5}\approx 16.9$*,* $C_{4.8}\approx 20.0$*), and* $|\mathcal{E}_{p;A,B} - A - q_pA\lambda| \le \frac12 C_pB^{p/2}$*. Since* $B^{p/2} = A\lambda^{(p-2)/2} = o(A\lambda)$ *as* $\lambda\to 0$*, the conjectured Poisson value is correct to second order as* $B^{p/2}/A\to 0$*, while the Gaussian-plus-far-jump value* $A + m_pB^{p/2}$ *misses the term* $q_pA\lambda$*. The bound is informative only in this rare-jump regime:* C_p *ranges over* $[12, 22.05]$ *on* $(4, 5]$*, about* 3.5–4 *times* m_p*.*

Near the boundary $B^{p/2}/A = 0$ of the parameter domain the conjecture itself holds. Here, for a finite Borel measure G on $\mathbb{R}\setminus\{0\}$ with $\int x^2dG<\infty$, $\tilde X_G$ denotes the centred compound Poisson variable, $\mathbb{E}e^{it\tilde X_G} = \exp\int(e^{itx}-1-itx)G(dx)$; for instance $\tilde X_{\lambda\delta_c} = c(\Pi_\lambda-\lambda)$.

Theorem I. *Let* $4<p<5$*. There is* $\lambda_0(p)>0$ *such that* $\mathcal{E}_{p;A,B} = c^p\mathbb{E}|\Pi_\lambda-\lambda|^p$ *for all* $A,B>0$ *with* $\lambda = (B^{p/2}/A)^{2/(p-2)} \le \lambda_0(p)$*, that is, with* $B^{p/2}\le \lambda_0(p)^{(p-2)/2}A$*. Moreover, for such* A, B *and every finite Borel measure* $G\notin\{\lambda\delta_c, \lambda\delta_{-c}\}$ *on* $\mathbb{R}\setminus\{0\}$ *with* $\int x^2dG = B$ *and* $\int|x|^pdG = A$*, one has* $\mathbb{E}|\tilde X_G|^p < c^p\mathbb{E}|\Pi_\lambda-\lambda|^p$*.*

The proof (Section 9) is perturbative: a quantitative stability estimate for the pair-interaction term of Proposition 9.6 is played off against a remainder that the two moment constraints make small to the same order. It yields an explicit $\lambda_0(p)$, but an astronomically small one (about $4\cdot10^{-28}$ at $p=4.5$ and $4\cdot10^{-17}$ at $p=4.8$; see Remark 9.9). The numerical evidence in Section 9 is consistent with the conjecture for all λ.

At the other end, $B^{p/2}/A\to\infty$, the conjectured value is confirmed to the first order beyond the Gaussian term:

Theorem J. *Let* $4<p<5$ *and* $K_p := p(p-1)(p-2)\,m_p/72$*. There is* $\tilde C_p<\infty$ *such that for all* $A,B>0$ *with* $\lambda = (B^{p/2}/A)^{2/(p-2)}\ge 1$*,*

$$c^p\mathbb{E}|\Pi_\lambda-\lambda|^p \le \mathcal{E}_{p;A,B} \le c^p\mathbb{E}|\Pi_\lambda-\lambda|^p + \tilde C_p\,B^{p/2}\lambda^{-2},$$
$$\Big|c^p\mathbb{E}|\Pi_\lambda-\lambda|^p - B^{p/2}\big(m_p + K_p\lambda^{-1}\big)\Big| \le \tilde C_p\,B^{p/2}\lambda^{-2}.$$

Hence $\mathcal{E}_{p;A,B} = B^{p/2}\big(m_p + K_p\lambda^{-1} + O(\lambda^{-2})\big)$ *as* $\lambda\to\infty$*. The Gaussian-plus-far-jump value* $A + m_pB^{p/2} = B^{p/2}(m_p+\lambda^{-(p-2)/2})$ *misses the term* $K_pB^{p/2}/\lambda$*.*

The proof rests on a second exact decomposition of $\mathbb{E}|\tilde X_G|^p$. Its leading part is *linear* in G and, for every λ, is maximized exactly by the measures carried by $\{\pm c\}$, by the concavity of an explicit function (Lemma 9.12). This is the two-constraint moment problem of [3] in its simplest instance, a planar curve. The constant K_p is the Edgeworth coefficient $\frac{(p)_4}{24}m_{p-4} + \frac{(p)_6}{72}m_{p-6}$, with $(p)_k = p(p-1)\cdots(p-k+1)$ and $m_s = 2^{s/2}\Gamma(\frac{s+1}{2})/\sqrt\pi$ continued analytically, since $m_s = (s-1)m_{s-2}$. For instance $K_4=1$ and $K_6=25$ are the coefficients of λ in $\mathbb{E}(\Pi_\lambda-\lambda)^4 = \lambda+3\lambda^2$ and of λ^2 in $\mathbb{E}(\Pi_\lambda-\lambda)^6 = \lambda+25\lambda^2+15\lambda^3$, and $K_{4.5} = 2.36055\ldots$. The second display is therefore an expansion of Edgeworth type (compare the classical expansions of absolute moments of normalized sums, e.g. [12, 2]); the proof below is self-contained. What is new in Theorem J is the upper bound for $\mathcal{E}_{p;A,B}$, uniform over all admissible summands, and Lemma 9.12. The proof works verbatim for $5\le p<6$, where the first display is weaker than Pinelis' exact result.

A stability analysis of the same decomposition gives exactness in the Gaussian regime as well:

Theorem K. *Let* $4<p<5$*. There is* $\lambda_1(p)<\infty$ *such that* $\mathcal{E}_{p;A,B} = c^p\mathbb{E}|\Pi_\lambda-\lambda|^p$ *for all* $A,B>0$ *with* $\lambda\ge\lambda_1(p)$*, that is, with* $B^{p/2}\ge\lambda_1(p)^{(p-2)/2}A$*. Moreover, for such* A, B *the last assertion of Theorem I holds:* $\mathbb{E}|\tilde X_G|^p < c^p\mathbb{E}|\Pi_\lambda-\lambda|^p$ *for every other admissible* G*. Consequently, Pinelis' conjecture holds for all* $A,B>0$ *with* $\lambda\notin(\lambda_0(p),\lambda_1(p))$*.*

The threshold $\lambda_1(p)$ produced by the proof is about 10^{12} at $p=4.5$, and between $10^{11.4}$ and 10^{13} for $p\in[4.05, 4.8]$. Unlike $\lambda_0(p)$ it stays bounded as $p\downarrow 4$, but it is far from optimal (Remark 9.17).

Remark 1.1 (Context). For $p = 4$, (1) follows from the identity $\mathbb{E}S^4 = \sum \mathbb{E}X_i^4 + 3(\sum \sigma_i^2)^2 - 3\sum \sigma_i^4$. For $p \geq 5$ and $p \in \{4, 6, 8, \dots\}$ the extremal value is $c^p \mathbb{E}|\Pi_\lambda - \lambda|^p$ (Poisson) [7, 1, 6]. For $p \in (4, 5)$, Pinelis' Poisson conjecture remains open outside the two extreme regimes of Theorems I and K. That $\mathcal{E}_{p;A,B} \geq A + m_p B^{p/2}$ holds for all $2 < p < 6$ follows from the construction in Section 3; Proposition G adds that for $4 < p < 6$ every zero-mean compound Poisson law lies above this value. The pointwise comparison inside a Fourier integral that proves Theorem A does not by itself reach $(4, 5)$. Theorems H and I combine the same identity with a separate extremal property of the Poisson law for the pair-interaction term (Proposition 9.6) and, for Theorem I, its stability (Lemma 9.7). Theorems J and K use a second decomposition adapted to the Gaussian regime (Lemmas 9.11–9.16). See Section 10.

2. Three lemmas

Lemma 2.1 (Fourier representation)**.** *Let $2 < p < 4$ and $c_p := \int_0^\infty (\cos s - 1 + \frac{s^2}{2}) s^{-p-1}\, ds$. Then $0 < c_p < \infty$ and for every $x \in \mathbb{R}$*

$$|x|^p = \frac{1}{c_p} \int_0^\infty \Big(\cos(tx) - 1 + \frac{t^2 x^2}{2} \Big) t^{-p-1}\, dt. \tag{2}$$

Consequently, for every random variable Y with $\mathbb{E}Y^2 < \infty$, writing $\varphi_Y(t) = \mathbb{E}e^{itY}$,

$$\mathbb{E}|Y|^p = \frac{1}{c_p} \int_0^\infty \Big(\operatorname{Re} \varphi_Y(t) - 1 + \frac{t^2 \mathbb{E}Y^2}{2} \Big) t^{-p-1}\, dt \in [0, \infty]. \tag{3}$$

Proof. Let $k(s) := \cos s - 1 + s^2/2$. Integrating $\sin s \leq s$ $(s \geq 0)$ three times from 0 gives $1 - \cos s \leq s^2/2$, $\sin s \geq s - s^3/6$ and $1 - \cos s \geq s^2/2 - s^4/24$; by evenness $0 \leq k(s) \leq s^4/24$ for all $s \in \mathbb{R}$, and $k(s) > 0$ for $s \neq 0$. Hence $k(s)s^{-p-1} \leq s^{3-p}/24$ is integrable near 0 because $p < 4$. For $s \geq 1$, $k(s) \leq 2 + s^2/2 \leq \frac{5}{2}s^2$, so $k(s)s^{-p-1} \leq \frac{5}{2}s^{1-p}$ is integrable at ∞ because $p > 2$. Hence $0 < c_p < \infty$. For $x \neq 0$ substitute $s = t|x|$ (cosine is even) to get $\int_0^\infty k(tx)t^{-p-1}dt = |x|^p c_p$; for $x = 0$ both sides of (2) vanish. For (3), apply (2) with $x = Y$ and take expectations. Since the integrand $k(tY)t^{-p-1}$ is nonnegative, Tonelli's theorem allows us to exchange $\mathbb{E}$ and $\int dt$, and $\mathbb{E}k(tY) = \operatorname{Re}\varphi_Y(t) - 1 + t^2\mathbb{E}Y^2/2$ because $\mathbb{E}Y^2 < \infty$. □

Definition 2.2. For a finite nonnegative Borel measure G on $\mathbb{R} \setminus \{0\}$, let X_G denote a random variable with $\mathbb{E}e^{itX_G} = \exp \int (e^{itx} - 1)G(dx)$, i.e. $X_G = \sum_{k=1}^N \xi_k$ with $N \sim \text{Poisson}(G(\mathbb{R}))$ and ξ_k i.i.d. with law $G/G(\mathbb{R})$, independent of N (and $X_G = 0$ if $G = 0$).

Lemma 2.3 (Compound Poisson facts)**.** *Let G be as above with $\int |x|^p G(dx) < \infty$ for some $p \geq 2$ and $\int x\, G(dx) = 0$. Then $\mathbb{E}|X_G|^p < \infty$, $\mathbb{E}X_G = 0$, $\mathbb{E}X_G^2 = \int x^2 G(dx)$, and*

$$\mathbb{E}e^{itX_G} = e^{-a(t)+ib(t)}, \quad a(t) := \int (1 - \cos tx) G(dx) \geq 0, \quad b(t) := \int \sin(tx)\, G(dx).$$

In particular $\operatorname{Re}\mathbb{E}e^{itX_G} = e^{-a(t)} \cos b(t)$. Moreover $a(t) \leq \frac{t^2}{2} \int x^2 G(dx)$. If $G_1, \dots, G_n$ are such measures and $X_{G_1}, \dots, X_{G_n}$ are independent, then $X_{G_1} + \cdots + X_{G_n} \stackrel{d}{=} X_{G_1 + \cdots + G_n}$.

Proof. Let $q = G(\mathbb{R}) > 0$ and $\xi \sim G/q$. Then $\mathbb{E}|\xi|^p < \infty$, and by convexity of $|\cdot|^p$, $|\sum_{k \leq N} \xi_k|^p \leq N^{p-1} \sum_{k \leq N} |\xi_k|^p$, whence $\mathbb{E}|X_G|^p \leq \mathbb{E}N^p\, \mathbb{E}|\xi|^p < \infty$. Conditioning on N gives $\mathbb{E}e^{itX_G} = \mathbb{E}(\varphi_\xi(t))^N = \exp(q(\varphi_\xi(t) - 1)) = \exp \int (e^{itx} - 1)G(dx)$, which has the stated real and imaginary parts. Conditioning on N and using $\mathbb{E}\xi = q^{-1} \int x\, dG = 0$: $\mathbb{E}X_G = \mathbb{E}N\, \mathbb{E}\xi = 0$ and $\mathbb{E}X_G^2 = \mathbb{E}N\, \mathbb{E}\xi^2 = \int x^2 dG$. The bound on a follows from $1 - \cos y \leq y^2/2$. The last claim holds because characteristic functions multiply under independent sums and the exponents add. □

Lemma 2.4 (Poissonization; Utev [11], see also [8] and [7, Thm. A])**.** *Let $p \geq 2$ and let $h \in C^2(\mathbb{R})$ with h'' convex and, for some constant C, $|h(x)| \leq C(1+|x|^p)$ and $|h''(x)| \leq C(1+|x|^{p-2})$ for all x. Let X be a random variable with $\mathbb{E}X = 0$ and $\mathbb{E}|X|^p < \infty$, and let G_X be the restriction of the law of X to $\mathbb{R} \setminus \{0\}$. Then $\mathbb{E}h(X) \leq \mathbb{E}h(X_{G_X})$.*

We include the short proof for completeness.

Proof. Let $q = \mathbb{P}(X \neq 0)$. If $q = 0$ both sides equal $h(0)$. Otherwise let $\xi, \xi_1, \xi_2, \dots$ be i.i.d. with law $\nu := G_X/q$ (the law of X given $X \neq 0$), $S_0 := 0$, $S_k := \xi_1 + \cdots + \xi_k$, and $N \sim \mathrm{Poisson}(q)$ independent, so that $X_{G_X} \overset{d}{=} S_N$. Note $\mathbb{E}\xi = \mathbb{E}[X\mathbf{1}_{X\neq 0}]/q = \mathbb{E}X/q = 0$ and $\mathbb{E}|\xi|^p = \mathbb{E}|X|^p/q < \infty$.

Step 1: $u(s) := \mathbb{E}h(s+\xi) - h(s)$ is convex. Integrating the bound on h'' gives $|h'(x)| \leq |h'(0)| + C(|x| + |x|^{p-1}) \leq C'(1+|x|^{p-1})$, and for s in a bounded interval I the random variables $|h(s+\xi)|$, $|h'(s+\xi)|$ and $|h''(s+\xi)|$ are dominated by $C''(1+|\xi|^p)$, which is integrable. (We use $|a+b|^r \leq 2^{(r-1)_+}(|a|^r + |b|^r)$ and $|\xi|^r \leq 1 + |\xi|^p$ for $0 \leq r \leq p$.) Hence we may differentiate twice under the expectation, so $u \in C^2$ and $u''(s) = \mathbb{E}h''(s+\xi) - h''(s)$. Since h'' is convex and $\mathbb{E}\xi = 0$, Jensen's inequality gives $\mathbb{E}h''(s+\xi) \geq h''(s)$. Thus $u'' \geq 0$.

Step 2: $\mathbb{E}h(S_k) \geq h(0) + k\,u(0)$ for all $k \geq 0$. The case $k = 0$ is trivial. For $k \geq 0$, independence of ξ_{k+1} and S_k and Fubini give $\mathbb{E}h(S_{k+1}) - \mathbb{E}h(S_k) = \mathbb{E}u(S_k)$; all expectations are finite since $\mathbb{E}|S_k|^p \leq k^p\mathbb{E}|\xi|^p$. By Jensen's inequality for the convex u and $\mathbb{E}S_k = 0$, $\mathbb{E}u(S_k) \geq u(0)$. Induct on k.

Step 3. Since $\sum_k \mathbb{P}(N=k)\mathbb{E}|h(S_k)| \leq C\sum_k \mathbb{P}(N=k)(1+k^p\mathbb{E}|\xi|^p) < \infty$,

$$\mathbb{E}h(S_N) = \sum_{k\geq 0} \mathbb{P}(N=k)\mathbb{E}h(S_k) \geq \sum_{k\geq 0} \mathbb{P}(N=k)\big(h(0)+ku(0)\big) = h(0) + q\big(\mathbb{E}h(\xi) - h(0)\big) = \mathbb{E}h(X),$$

because the law of X is $(1-q)\delta_0 + q\nu$. □

Corollary 2.5. *Let $p \geq 3$, let W be a random variable with $\mathbb{E}|W|^p < \infty$, and let $W, X_1, \dots, X_n$ be independent with $\mathbb{E}X_i = 0$, $\mathbb{E}|X_i|^p < \infty$. Put $G := \sum_i G_{X_i}$ and let X_G be independent of W. Then $\mathbb{E}|W + X_1 + \cdots + X_n|^p \leq \mathbb{E}|W + X_G|^p$.*

Proof. Let $X_{G_{X_1}}, \dots, X_{G_{X_n}}$ be independent compound Poisson variables, independent of everything else. We replace $X_1, \dots, X_n$ one at a time. Suppose $X_1, \dots, X_{j-1}$ have been replaced, and let $V := W + \sum_{i<j} X_{G_{X_i}} + \sum_{i>j} X_i$. Then V is independent of X_j and of $X_{G_{X_j}}$, and $\mathbb{E}|V|^p < \infty$ by Lemma 2.3. For fixed v, the function $h_v(x) := |v+x|^p$ is C^2 with $h_v''(x) = p(p-1)|v+x|^{p-2}$, which is convex because $p-2 \geq 1$, and it satisfies the growth bounds of Lemma 2.4 with $C = C_p(1+|v|^p)$. Lemma 2.4 gives $\mathbb{E}h_v(X_j) \leq \mathbb{E}h_v(X_{G_{X_j}})$. Integrating over the law of V (Fubini; both sides are finite) yields $\mathbb{E}|V + X_j|^p \leq \mathbb{E}|V + X_{G_{X_j}}|^p$. After n steps, use $\sum_i X_{G_{X_i}} \overset{d}{=} X_G$ (Lemma 2.3). □

3. Proof of Theorem A

The case $p = 4$. Expanding, independence and $\mathbb{E}X_i = 0$ give $\mathbb{E}S^4 = \sum_i \mathbb{E}X_i^4 + 6\sum_{i<j}\mathbb{E}X_i^2\mathbb{E}X_j^2 = A + 3B^2 - 3\sum_i(\mathbb{E}X_i^2)^2 \leq A + m_4B^2$, since $m_4 = 3$.

The upper bound for $3 \leq p < 4$. Let $X_1, \dots, X_n$ be as in the theorem, $G = \sum_i G_{X_i}$. Then G is a finite measure on $\mathbb{R}\setminus\{0\}$ with $\int x\,dG = \sum \mathbb{E}X_i = 0$, $\int x^2 dG = B$ and $\int |x|^p dG = A$. By Corollary 2.5 with $W = 0$, $\mathbb{E}|S|^p \leq \mathbb{E}|X_G|^p$. It remains to show that

$$\mathbb{E}|X_G|^p \leq A + m_p B^{p/2} \qquad (2 < p < 4). \tag{4}$$

Write $g(t) := t^2B/2$ and let a, b be as in Lemma 2.3. Each of the three quantities in (4) is finite. Lemma 2.1 applies to each of them:

$$c_p\,\mathbb{E}|X_G|^p = \int_0^\infty \big(e^{-a(t)}\cos b(t) - 1 + g(t)\big)t^{-p-1}dt,$$
$$c_p\,A = \int_0^\infty \int \Big(\cos tx - 1 + \frac{t^2x^2}{2}\Big)G(dx)\,t^{-p-1}dt = \int_0^\infty \big(-a(t) + g(t)\big)t^{-p-1}dt,$$
$$c_p\,m_pB^{p/2} = c_p\mathbb{E}|B^{1/2}Z|^p = \int_0^\infty \big(e^{-g(t)} - 1 + g(t)\big)t^{-p-1}dt.$$

The first line is (3) combined with Lemma 2.3. In the second line we integrated (2) against G and used Tonelli for the nonnegative integrand. In the third we used $\mathbb{E}e^{itB^{1/2}Z} = e^{-g(t)}$. All three integrands are nonnegative and the integrals are finite, so we may subtract:

$$c_p\big(\mathbb{E}|X_G|^p - A - m_pB^{p/2}\big) = \int_0^\infty \Big(e^{-a(t)}\cos b(t) + a(t) - e^{-g(t)} - g(t)\Big)t^{-p-1}dt. \tag{5}$$

Let $F(x) := e^{-x} + x$. Then $F'(x) = 1 - e^{-x} \geq 0$ for $x \geq 0$, so F is nondecreasing on $[0,\infty)$. By Lemma 2.3, $0 \leq a(t) \leq g(t)$ for every t. Hence, pointwise in t,

$$e^{-a}\cos b + a - e^{-g} - g \;\leq\; F(a) - F(g) \;\leq 0.$$

Since $c_p > 0$, (5) yields (4), and hence (1).

Sharpness. Fix $A, B > 0$ and $2 < p \leq 4$. We construct $\mathbf{X}^{(n)} \in \mathscr{X}_{p;A,B}$ with $\mathbb{E}|S_{\mathbf{X}^{(n)}}|^p \to A + m_pB^{p/2}$. (This is essentially [7, Prop. 1.6]; we give a self-contained version with exact constraints.) Let $\chi(\varepsilon) := \varepsilon(1-\varepsilon)\big((1-\varepsilon)^{p-1} + \varepsilon^{p-1}\big)$, so that for $J = M(\beta - \varepsilon)$ with $\beta \sim$ Bernoulli(ε) we have $\mathbb{E}J = 0$, $\mathbb{E}J^2 = M^2\varepsilon(1-\varepsilon)$ and $\mathbb{E}|J|^p = M^p\chi(\varepsilon)$. Let $r_1, r_2, \ldots$ be i.i.d. Rademacher signs independent of β. For $n \in \mathbb{N}$, $\varepsilon \in (0, \frac{1}{2})$ and $0 \leq M \leq M_{\max} := (B/(\varepsilon(1-\varepsilon)))^{1/2}$ let $s := (B - M^2\varepsilon(1-\varepsilon))^{1/2}$ and

$$\mathbf{X} = (J,\ sn^{-1/2}r_1, \ldots, sn^{-1/2}r_n), \qquad \sum \mathbb{E}X_i^2 = B, \quad \sum \mathbb{E}|X_i|^p = \phi(M) := M^p\chi(\varepsilon) + s^pn^{1-p/2}.$$

Now ϕ is continuous on $[0, M_{\max}]$, with $\phi(0) = B^{p/2}n^{1-p/2}$ and $\phi(M_{\max}) = M_{\max}^p\chi(\varepsilon) \geq (B/\varepsilon)^{p/2}\varepsilon(1-\varepsilon)^p \geq B^{p/2}2^{-p}\varepsilon^{1-p/2}$. Choose n so large that $B^{p/2}n^{1-p/2} < A$, and then $\varepsilon = \varepsilon_n \in (0, 1/n)$ so small that $\phi(M_{\max}) > A$. By the intermediate value theorem there is $M = M_n$ with $\phi(M_n) = A$, so $\mathbf{X}^{(n)} \in \mathscr{X}_{p;A,B}$. As $n \to \infty$: $M_n^p\chi(\varepsilon_n) = A - s^pn^{1-p/2} \to A$. Since $\chi(\varepsilon)/\varepsilon \to 1$, also $M_n^p\varepsilon_n \to A$, whence $M_n \to \infty$, $M_n\varepsilon_n = (M_n^p\varepsilon_n)^{1/p}\varepsilon_n^{1-1/p} \to 0$ and $M_n^2\varepsilon_n = (M_n^p\varepsilon_n)^{2/p}\varepsilon_n^{1-2/p} \to 0$, so $s = s_n \to B^{1/2}$. Let $R_n := s_nn^{-1/2}\sum_{i\leq n} r_i$. Then $R_n \to B^{1/2}Z$ in distribution (CLT) and $\mathbb{E}R_n^6 \leq 15s_n^6$, so $(|R_n|^p)$ is uniformly integrable ($p \leq 4$) and $\|R_n\|_p \to m_p^{1/p}B^{1/2}$. Conditioning on β,

$$\mathbb{E}|S|^p = (1-\varepsilon_n)\mathbb{E}|R_n - M_n\varepsilon_n|^p + \varepsilon_nM_n^p\,\mathbb{E}\big|(1-\varepsilon_n) + R_n/M_n\big|^p.$$

By Minkowski, $|\|R_n - M_n\varepsilon_n\|_p - \|R_n\|_p| \leq M_n\varepsilon_n \to 0$ and $|\|(1-\varepsilon_n) + R_n/M_n\|_p - (1-\varepsilon_n)| \leq \|R_n\|_p/M_n \to 0$. Hence $\mathbb{E}|S|^p \to m_pB^{p/2} + A$. Together with (1) this proves $\mathcal{E}_{p;A,B} = A + m_pB^{p/2}$ for $3 \leq p \leq 4$. □

4. Proof of Corollary B

Finite dimension. Let $\mathbb{H} = \mathbb{R}^d$ and let σ be the normalized surface measure on S^{d-1}. By rotation invariance, $\int_{S^{d-1}} |\langle\theta, x\rangle|^p\sigma(d\theta) = \gamma_{d,p}\|x\|^p$ for all x, with $\gamma_{d,p} := \int |\theta_1|^p\sigma(d\theta) \in (0,\infty)$. For each fixed θ the real random variables $\langle\theta, X_i\rangle$ are independent, have mean zero, satisfy $\mathbb{E}|\langle\theta, X_i\rangle|^p \leq$

$\mathbb{E}\|X_i\|^p < \infty$, and $\sum_i \mathbb{E}\langle\theta, X_i\rangle^2 = \langle\Sigma\theta,\theta\rangle$, where Σ is the covariance matrix of Γ. By Theorem A for $3 \le p \le 4$, and by [7, Thm. 1.5] with $X = 0$ for $2 < p \le 3$,

$$\mathbb{E}|\langle\theta, S\rangle|^p \le \sum_i \mathbb{E}|\langle\theta, X_i\rangle|^p + m_p\langle\Sigma\theta,\theta\rangle^{p/2} = \sum_i \mathbb{E}|\langle\theta, X_i\rangle|^p + \mathbb{E}|\langle\theta,\Gamma\rangle|^p,$$

because $\langle\theta,\Gamma\rangle \sim N(0, \langle\Sigma\theta,\theta\rangle)$. Integrating in θ against σ (Tonelli) and dividing by $\gamma_{d,p}$ gives the first inequality.

Separable $\mathbb{H}$. Let P_N be the orthogonal projection onto the span of the first N vectors of an orthonormal basis. The vectors $P_N X_i$ are independent and zero-mean in an N-dimensional space, and $P_N\Gamma$ is centered Gaussian with covariance $P_N\Sigma P_N = \sum_i \mathrm{Cov}(P_N X_i)$. Hence $\mathbb{E}\|P_N S\|^p \le \sum_i \mathbb{E}\|P_N X_i\|^p + \mathbb{E}\|P_N\Gamma\|^p \le \sum_i \mathbb{E}\|X_i\|^p + \mathbb{E}\|\Gamma\|^p$. Since $\|P_N S\| \uparrow \|S\|$, monotone convergence gives the first inequality.

Second inequality. Write $\Gamma = \sum_j \sqrt{\lambda_j} Z_j e_j$ (Karhunen–Loève), with Z_j i.i.d. $N(0,1)$ and $\sum_j \lambda_j = \mathbb{E}\|\Gamma\|^2 = \sum_i \mathbb{E}\|X_i\|^2$. Minkowski's inequality in $L^{p/2}$ ($p/2 \ge 1$) gives $\| \|\Gamma\|^2\|_{p/2} \le \sum_j \lambda_j \|Z_j^2\|_{p/2} = m_p^{2/p} \sum_j \lambda_j$, i.e. $\mathbb{E}\|\Gamma\|^p \le m_p(\sum_i \mathbb{E}\|X_i\|^2)^{p/2}$. The constants are optimal already for $\mathbb{H} = \mathbb{R}$ (Theorem A and [7, Thm. 1.5]). □

5. Proof of Theorem D

Let $\varphi_X(t) := \mathbb{E}e^{itX} \in [0,1]$; φ_X is real, so X is symmetric, $\mathbb{E}X = 0$ and $\mathbb{E}X^2 < \infty$. For $p = 4$: $\mathbb{E}(X+S)^4 = \mathbb{E}X^4 + 6\mathbb{E}X^2\,\mathbb{E}S^2 + \mathbb{E}S^4 \le \mathbb{E}X^4 + 6B\,\mathbb{E}X^2 + A + 3B^2 = A + \mathbb{E}(X + B^{1/2}Z)^4$, using $\mathbb{E}X = \mathbb{E}S = 0$ and the case $p = 4$ of Theorem A. Let now $3 \le p < 4$. Corollary 2.5 with $W = X$ gives $\mathbb{E}|X+S|^p \le \mathbb{E}|X+X_G|^p$ with $G = \sum_i G_{X_i}$. By (3), using $\mathbb{E}(X+X_G)^2 = \mathbb{E}(X+B^{1/2}Z)^2 = \mathbb{E}X^2 + B$ and $\mathrm{Re}(\varphi_X(t)e^{-a+ib}) = \varphi_X(t)e^{-a}\cos b$, the same computation as in (5) gives

$$c_p\big(\mathbb{E}|X + X_G|^p - A - \mathbb{E}|X + B^{1/2}Z|^p\big) = \int_0^\infty \Big(\varphi_X\left(e^{-a}\cos b - e^{-g}\right) + a - g\Big)t^{-p-1}dt.$$

Since $0 \le \varphi_X \le 1$ and $e^{-a} - e^{-g} \ge 0$, we get $\varphi_X(e^{-a}\cos b - e^{-g}) \le \varphi_X(e^{-a} - e^{-g}) \le e^{-a} - e^{-g}$, so the integrand is $\le F(a) - F(g) \le 0$. For sharpness, the sequence of the previous section satisfies $\mathbb{E}|X + S_{\mathbf{X}^{(n)}}|^p \to A + \mathbb{E}|X + B^{1/2}Z|^p$. Indeed, the same conditioning on β gives $(1-\varepsilon_n)\mathbb{E}|X + R_n - M_n\varepsilon_n|^p + \varepsilon_n M_n^p\mathbb{E}|(1-\varepsilon_n) + (X+R_n)/M_n|^p$, and $X + R_n \to X + B^{1/2}Z$ in distribution with uniformly integrable p-th powers. □

6. Proof of Theorem E

(a). It suffices to treat $x_0 = 1$: if X_1 works for $x_0 = 1$, then $x_0 X_1$ (a zero-mean two-point variable) works for x_0, by homogeneity of both sides and the symmetry of Z when $x_0 < 0$. Fix $u \in (0,1)$ and $\lambda \in (0, \frac{1}{2}]$, and let $X_1 := u(\beta - \lambda)$ with $\beta \sim \mathrm{Bernoulli}(\lambda)$. Then

$$L := \mathbb{E}|1+X_1|^p = (1-\lambda)(1-u\lambda)^p + \lambda(1+u-u\lambda)^p, \quad A = u^p\big(\lambda(1-\lambda)^p + (1-\lambda)\lambda^p\big), \quad B = u^2\lambda(1-\lambda).$$

Lower bound for L. By Bernoulli's inequality $(1-u\lambda)^p \ge 1 - pu\lambda$. By convexity of $y \mapsto y^p$, $(1+u-u\lambda)^p \ge (1+u)^p - pu\lambda(1+u)^{p-1}$. Hence

$$L \ge 1 + \lambda\big((1+u)^p - 1 - pu\big) - pu(1+u)^{p-1}\lambda^2.$$

Upper bound for $R := A + \mathbb{E}|1 + B^{1/2}Z|^p$. We claim there is $\Theta_p < \infty$ with

$$|1+y|^p \le 1 + py + \tbinom{p}{2}y^2 + \tbinom{p}{3}y^3 + \Theta_p y^4 \qquad (y \in \mathbb{R}). \tag{6}$$

For $y \ge -\frac{1}{2}$, Taylor's formula with Lagrange remainder gives remainder $\binom{p}{4}(1+\theta y)^{p-4}y^4$ for some $\theta \in (0,1)$, and $(1+\theta y)^{p-4} \le 2^{4-p} \le 2$. For $y < -\frac{1}{2}$ we have $|y|^k \le 2^{4-k}y^4$ for $k \le 4$, $|1+y|^p \le (3|y|)^p \le 3^p 2^{4-p}y^4$, and $-(1 + py + \binom{p}{2}y^2 + \binom{p}{3}y^3) \le (16 + 8p + 4\binom{p}{2} + 2\binom{p}{3})y^4$. This proves

the claim. Taking $y = B^{1/2}Z$ in (6) and using $\mathbb{E}Z = \mathbb{E}Z^3 = 0$, $\mathbb{E}Z^4 = 3$, we get $\mathbb{E}|1 + B^{1/2}Z|^p \le 1 + \binom{p}{2}B + 3\Theta_p B^2$. With $A \le u^p\lambda + u^p\lambda^p$ and $B \le u^2\lambda$:

$$R \le 1 + \lambda\big(u^p + \tbinom{p}{2}u^2\big) + u^p\lambda^p + 3\Theta_p u^4\lambda^2.$$

Conclusion. Let $D(u) := (1+u)^p - 1 - pu - \binom{p}{2}u^2 - u^p$. Then

$$L - R \ge \lambda D(u) - \big(pu(1+u)^{p-1} + 3\Theta_p u^4\big)\lambda^2 - u^p\lambda^p.$$

By Taylor's formula at order 4 (remainder $\binom{p}{4}(1+\theta u)^{p-4}u^4 \ge 0$, since $\binom{p}{4} > 0$ for $3 < p < 4$), $D(u) \ge \binom{p}{3}u^3 - u^p$. Moreover $\binom{p}{3} = \frac{p(p-1)(p-2)}{6} > 1$ and $u^{p-3} < 1$, so $D(u) > u^3(1 - u^{p-3}) > 0$. Since $p > 2$, the right-hand side above is positive for all sufficiently small $\lambda > 0$. Thus $L > R$.

(b). Let $|X| \le K$ and $|X| \ge \delta$ a.s., with $0 < \delta \le K < \infty$. Put $f(x) = |x|^p$, $\tau := \mathbb{E}f'''(X) = p(p-1)(p-2)\mathbb{E}[|X|^{p-3}\operatorname{sign} X] \ne 0$, and $\sigma := \operatorname{sign}\tau$. For $0 < u \le \delta/4$ and $0 < \lambda \le \frac14$ let $X_1 := \sigma u(\beta - \lambda)$ with $\beta \sim \text{Bernoulli}(\lambda)$ independent of X. Then $\mathbb{E}X_1 = 0$, $B := \mathbb{E}X_1^2 = u^2\lambda(1-\lambda)$ and $\mathbb{E}|X_1|^p \le u^p(\lambda + \lambda^p)$.

Two Taylor bounds. Let $|x| \in [\delta, K]$. On $\{|y| \le \delta/2\}$ the function $y \mapsto f(x+y)$ is C^∞. Moreover $|f''''(z)| = p(p-1)(p-2)(p-3)|z|^{p-4} \le p(p-1)(p-2)(p-3)(\delta/2)^{p-4}$ for $|z| \ge \delta/2$. Hence, with $C_1 := \binom{p}{4}(\delta/2)^{p-4}$ and $T_x(y) := f(x) + f'(x)y + \frac12 f''(x)y^2 + \frac16 f'''(x)y^3$,

$$|f(x+y) - T_x(y)| \le C_1 y^4 \qquad (|y| \le \delta/2). \tag{7}$$

For $|y| > \delta/2$ we have $K \le (2K/\delta)|y|$, so $f(x+y) \le (K + |y|)^p \le (1 + 2K/\delta)^p|y|^p \le (1 + 2K/\delta)^p(2/\delta)^{4-p}y^4$. Also $|T_x(y)| \le \sum_{k=0}^3 |f^{(k)}(x)||y|^k/k! \le C_2' y^4$ with $C_2' := \sum_{k=0}^3 \frac{|f^{(k)}|_{\max}}{k!}(2/\delta)^{4-k}$, where $|f^{(k)}|_{\max} := \max_{\delta \le |z| \le K}|f^{(k)}(z)|$. Therefore, with $C_2 := \max\{C_1, (1+2K/\delta)^p(2/\delta)^{4-p} + C_2'\}$,

$$f(x+y) \le T_x(y) + C_2 y^4 \qquad (y \in \mathbb{R}). \tag{8}$$

Lower bound for $L := \mathbb{E}|X + X_1|^p$. Condition on $X = x$. The two values of X_1 are $y_1 = -\sigma u\lambda$ and $y_2 = \sigma u(1-\lambda)$, both of modulus $\le u \le \delta/4$. Using (7) and the identities $(1-\lambda)y_1 + \lambda y_2 = 0$, $(1-\lambda)y_1^2 + \lambda y_2^2 = B$ and $(1-\lambda)y_1^3 + \lambda y_2^3 = \sigma u^3\lambda(1-\lambda)(1-2\lambda)$,

$$(1-\lambda)f(x+y_1) + \lambda f(x+y_2) \ge f(x) + \tfrac12 f''(x)B + \tfrac16 f'''(x)\,\sigma u^3\lambda(1-\lambda)(1-2\lambda) - C_1u^4\lambda\big(\lambda^3 + (1-\lambda)^4\big).$$

Taking expectation in x and using $\sigma\tau = |\tau|$, $\lambda^3 + (1-\lambda)^4 \le 2$:

$$L \ge \mathbb{E}f(X) + \tfrac12 B\,\mathbb{E}f''(X) + \tfrac16|\tau|u^3\lambda(1-\lambda)(1-2\lambda) - 2C_1u^4\lambda.$$

Upper bound for $R := \mathbb{E}|X_1|^p + \mathbb{E}|X + B^{1/2}Z|^p$. By (8) with $y = B^{1/2}Z$, using $\mathbb{E}Z = \mathbb{E}Z^3 = 0$ and $\mathbb{E}Z^4 = 3$, $\mathbb{E}f(x + B^{1/2}Z) \le f(x) + \frac12 f''(x)B + 3C_2B^2$. Hence

$$R \le u^p(\lambda + \lambda^p) + \mathbb{E}f(X) + \tfrac12 B\,\mathbb{E}f''(X) + 3C_2u^4\lambda^2.$$

Conclusion. Since $(1-\lambda)(1-2\lambda) \ge \frac38$ for $\lambda \le \frac14$,

$$\frac{L-R}{\lambda} \;\ge\; \frac{|\tau|}{16}u^3 - 2C_1u^4 - u^p - u^p\lambda^{p-1} - 3C_2u^4\lambda.$$

Because $p > 3$, we can first fix $u \in (0, \delta/4]$ with $\frac{|\tau|}{16}u^3 > 2(2C_1u^4 + u^p)$, and then $\lambda \in (0, \frac14]$ so small that $u^p\lambda^{p-1} + 3C_2u^4\lambda < 2C_1u^4 + u^p$. Then $L > R$. Part (a) is the special case $X \equiv x_0$; the separate proof of (a) above shows that there every $u \in (0,1)$ works (after scaling to $x_0 = 1$). □

Remark 6.1 (Numerical instances; 30-digit evaluation). The Gaussian terms were evaluated with $\mathbb{E}|x + \sigma Z|^p = \sigma^p 2^{p/2}\Gamma(\frac{p+1}{2})\pi^{-1/2}\,{}_1F_1(-\frac{p}{2}; \frac12; -\frac{x^2}{2\sigma^2})$. (i) In (a) with $p = 3.5$, $u = \frac12$, $\lambda = \frac{1}{10}$: $L = 1.119202\ldots$, $R = 1.104991\ldots$, $L - R = 0.01421\ldots$; the gap stays positive for $p = 3.05, 3.2, 3.8, 3.95$, being 0.0010, 0.0050, 0.0255 and 0.0321 respectively. (ii) In (b) with $X \in \{-1, 3\}$ with probabilities $\frac34, \frac14$ (mean zero), $p = 3.5$, $X_1 = -\frac12(\beta - \frac{1}{10})$: $L = 12.65025836\ldots$, $R = 12.64955547\ldots$ (this instance has $u = \frac12 > \delta/4$, so it is not produced by the recipe of the proof, and it is specific to the exponent:

with the same X_1 the gap is negative at $p = 3.2$ and at $p = 3.95$; the recipe of the proof, with smaller u, λ, works at every $p \in (3, 4)$). These are single zero-mean two-point summands. For $p \le 3$, Pinelis' Theorem 1.5 holds for every such configuration.

Remark 6.2 ($p = 4$). At $p = 4$ the extension still fails for non-centered X: for $X \equiv 1$ and any zero-mean X_1, $\mathbb{E}(1 + X_1)^4 - \mathbb{E}X_1^4 - \mathbb{E}(1 + B^{1/2}Z)^4 = 4\mathbb{E}X_1^3 - 3B^2$, which equals $0.0345\ldots$ for the instance (i). For centered X the extension holds at $p = 4$ (expand the fourth power), consistent with $\tau \to 0$ as $p \uparrow 4$ for centered X.

7. Proof of Theorem F

Regularity of f. Let $k(s) = \cos s - 1 + s^2/2$. Since $0 \le k(s) \le \min(s^4/24, s^2/2)$, we have $0 \le f(x) - \kappa x^4 \le (x^2 + x^4)\int (t^2 \wedge t^4)d\mu$. Moreover $|\partial_x k(tx)| = t|tx - \sin tx| \le t\min(|tx|^3/6, 2|tx|)$ and $0 \le \partial_x^2 k(tx) = t^2(1 - \cos tx) \le \min(t^4x^2/2, 2t^2)$. These are dominated, locally uniformly in x, by multiples of $(t^2 \wedge t^4)$. By dominated convergence $f \in C^2$, with

$$f''(x) = 12\kappa x^2 + \int t^2(1 - \cos tx)\,\mu(dt), \qquad 0 \le f''(x) \le C(1 + x^2), \qquad 0 \le f(x) \le C(1 + x^4).$$

Poissonization. For fixed v, $h_v(x) := f(v + x)$ satisfies the hypotheses of Lemma 2.4 with exponent $p = 4$ (convex h_v''; $|h_v| \le C_v(1 + x^4)$, $|h_v''| \le C_v(1 + x^2)$). Hence the proof of Corollary 2.5, with $|\cdot|^p$ replaced by f and p-th moments by fourth moments, gives $\mathbb{E}f(S) \le \mathbb{E}f(X_G)$ with $G = \sum_i G_{X_i}$. Here $\mathbb{E}X_G^4 < \infty$ by Lemma 2.3 with $p = 4$.

Fourier step. For any Y with $\mathbb{E}Y^4 < \infty$, Tonelli gives $\mathbb{E}f(Y) = \kappa\mathbb{E}Y^4 + \int(\operatorname{Re}\varphi_Y(t) - 1 + \frac{t^2}{2}\mathbb{E}Y^2)\mu(dt)$, all terms finite. For the quartic part, $\mathbb{E}X_G^4 = \int x^4 dG + 3B^2$ (the fourth cumulant of X_G is $\int x^4 dG$ and $\mathbb{E}X_G = 0$), so $\kappa\mathbb{E}X_G^4 = \kappa\int x^4 dG + \kappa\mathbb{E}(B^{1/2}Z)^4$. For the μ-part, exactly as in (5),

$$\int \big(\operatorname{Re}\varphi_{X_G} - 1 + g\big)d\mu - \iint k(tx)\,G(dx)\mu(dt) - \int \big(e^{-g} - 1 + g\big)d\mu = \int \big(e^{-a}\cos b + a - e^{-g} - g\big)d\mu \le 0.$$

Adding up, $\mathbb{E}f(X_G) \le \int f\,dG + \mathbb{E}f(B^{1/2}Z) = \sum_i \mathbb{E}f(X_i) + \mathbb{E}f(B^{1/2}Z)$. □

8. Proof of Proposition G

For $4 < p < 6$ let $k_4(s) := 1 - \cos s - \frac{s^2}{2} + \frac{s^4}{24}$. Continuing the integration of $\sin s \le s$ from the proof of Lemma 2.1 two more times gives $\sin s \le s - s^3/6 + s^5/120$ and then $1 - \cos s \le s^2/2 - s^4/24 + s^6/720$ for $s \ge 0$; together with $1 - \cos s \ge s^2/2 - s^4/24$ and evenness, $0 \le k_4(s) \le s^6/720$ for all s. Also $k_4(s) \le s^4/24 + 3$, so $c_p' := \int_0^\infty k_4(s)s^{-p-1}ds \in (0, \infty)$ and $|x|^p = \frac{1}{c_p'}\int_0^\infty k_4(tx)t^{-p-1}dt$ for all x. For Y with $\mathbb{E}Y^4 < \infty$, Tonelli gives $c_p'\mathbb{E}|Y|^p = \int_0^\infty (1 - \operatorname{Re}\varphi_Y - \frac{t^2}{2}\mathbb{E}Y^2 + \frac{t^4}{24}\mathbb{E}Y^4)t^{-p-1}dt$. For $Y = X_G$ one has $\mathbb{E}X_G^4 = \int x^4 dG + 3B^2$, by the cumulant formula, the fourth cumulant being $\int x^4 dG$. Also $\mathbb{E}|B^{1/2}Z|^4 = 3B^2$ and $\int k_4(tx)G(dx) = a - g + \frac{t^4}{24}\int x^4 dG$. The terms with $\int x^4 dG$ cancel, and

$$\begin{aligned} c_p'\big(\mathbb{E}|X_G|^p - A - m_p B^{p/2}\big) &= \int_0^\infty \big(F(g) - a - e^{-a}\cos b\big)t^{-p-1}dt \\ &\ge \int_0^\infty \big(F(g) - F(a)\big)t^{-p-1}dt \ge 0. \qquad \square \end{aligned}$$

9. Symmetrization, a second proof, and the range $4 < p < 5$

For a finite Borel measure G on $\mathbb{R} \setminus \{0\}$ with $\int x^2 dG < \infty$ let $\tilde{X}_G$ be the *centred* compound Poisson variable, $\mathbb{E}e^{it\tilde{X}_G} = \exp\int(e^{itx} - 1 - itx)G(dx)$. When $\int x\,dG = 0$, $\tilde{X}_G = X_G$. Let $G_s(E) := \frac{1}{2}(G(E) + G(-E))$ be the symmetrization of G. Then $\mathbb{E}e^{it\tilde{X}_G} = e^{-a(t) + ib(t)}$ with $a = \int(1 - \cos tx)dG$

and $b = \int(\sin tx - tx)dG$, and $\mathbb{E}e^{itX_{G_s}} = e^{-a(t)}$, with the same a. (If $\int x\,dG = 0$ this b agrees with that of Lemma 2.3; in general $\tilde{X}_G = X_G - \int x\,dG$.)

Proposition 9.1. *Let G be as above with $\int |x|^p dG < \infty$.*

(a) *If $2 < p < 4$, then $\mathbb{E}|\tilde{X}_G|^p \le \mathbb{E}|X_{G_s}|^p$.*
(b) *If $4 < p < 6$, then $\mathbb{E}|\tilde{X}_G|^p \ge \mathbb{E}|X_{G_s}|^p$, and with c'_p as in the proof of Proposition G,*

$$c'_p\big(\mathbb{E}|\tilde{X}_G|^p - \mathbb{E}|X_{G_s}|^p\big) = \int_0^\infty e^{-a(t)}\big(1-\cos b(t)\big)t^{-p-1}dt.$$

Proof. Both variables have mean 0 and variance $B = \int x^2 dG$, and for $p > 4$ both have fourth moment $\int x^4 dG + 3B^2$, since the fourth cumulants agree. (a) By (3), $c_p(\mathbb{E}|\tilde{X}_G|^p - \mathbb{E}|X_{G_s}|^p) = \int_0^\infty e^{-a}(\cos b - 1)t^{-p-1}dt \le 0$. (b) By the representation in the proof of Proposition G, $c'_p(\mathbb{E}|\tilde{X}_G|^p - \mathbb{E}|X_{G_s}|^p) = \int_0^\infty (e^{-a} - e^{-a}\cos b)t^{-p-1}dt \ge 0$. All integrals converge absolutely by the same bounds as before. □

Remark 9.2 (A second proof of Theorem A). For $3 \le p < 4$, Corollary 2.5, Proposition 9.1(a) and the symmetric exact bound ([4, Thm. 6.1 and Cor. 6.2], [5]) give $\mathbb{E}|S|^p \le \mathbb{E}|X_G|^p \le \mathbb{E}|X_{G_s}|^p \le A + m_p B^{p/2}$. For the last step, X_{G_s} is the distributional limit of $S_n = \sum_{j\le n} V_{j,n}$ with $V_{j,n}$ i.i.d. symmetric of law $(1 - G_s(\mathbb{R})/n)\delta_0 + G_s/n$. These satisfy $\sum_j \mathbb{E}V_{j,n}^2 = B$ and $\sum_j \mathbb{E}|V_{j,n}|^p = A$ exactly, so by Fatou's lemma for weak convergence $\mathbb{E}|X_{G_s}|^p \le \liminf_n \mathbb{E}|S_n|^p \le A + m_p B^{p/2}$. Thus for $3 \le p \le 4$ the zero-mean problem reduces to the symmetric one. For $4 < p < 6$ this reduction is impossible: for $p > 4$ the symmetric supremum is $c^p\mathbb{E}|\Pi_{\lambda/2} - \Pi'_{\lambda/2}|^p$ (Utev; [7, eq. (12)]), which by Proposition 9.1(b) applied to $G_* = \lambda\delta_c$ (the integrand $e^{-a}(1-\cos b)$ being positive on a set of positive measure) is strictly below $c^p\mathbb{E}|\Pi_\lambda - \lambda|^p$, a lower bound for $\mathcal{E}_{p;A,B}$ by [7, Lemma 2.1] and Fatou's lemma. Skewing the Lévy measure *increases* the p-th moment; this is why $p = 4$ is the threshold.

Proposition 9.3 (Conditional reduction for $4 < p < 5$)**.** *Let $4 < p < 5$ and $A, B > 0$; let c, λ be given by $c^p\lambda = A$, $c^2\lambda = B$, and $G_* = \lambda\delta_c$. Suppose that the following* skewness inequality *holds for every finite measure G on $\mathbb{R}\setminus\{0\}$ with $\int x\,dG = 0$, $\int x^2 dG = B$ and $\int |x|^p dG = A$:*

$$\mathbb{E}|\tilde{X}_G|^p - \mathbb{E}|X_{G_s}|^p \;\le\; \mathbb{E}|\tilde{X}_{G_*}|^p - \mathbb{E}|X_{(G_*)_s}|^p = c^p\big(\mathbb{E}|\Pi_\lambda - \lambda|^p - \mathbb{E}|\Pi_{\lambda/2} - \Pi'_{\lambda/2}|^p\big). \tag{9}$$

Then $\mathcal{E}_{p;A,B} = c^p\mathbb{E}|\Pi_\lambda - \lambda|^p$. In particular, if (9) *holds for all $A, B > 0$ (by homogeneity: for all $\lambda > 0$), then Pinelis' conjecture holds for this p.*

Proof. Let $\mathbf{X} \in \mathscr{X}_{p;A,B}$ and $G = \sum_i G_{X_i}$. Then $\int x\,dG = 0$, so $\tilde{X}_G = X_G$, and Corollary 2.5 gives $\mathbb{E}|S_{\mathbf{X}}|^p \le \mathbb{E}|X_G|^p$. Write $\mathbb{E}|X_G|^p = (\mathbb{E}|X_G|^p - \mathbb{E}|X_{G_s}|^p) + \mathbb{E}|X_{G_s}|^p$. The first bracket is at most the right-hand side of (9). For the second, Utev's symmetric theorem [11] (see [7, eq. (12)]), valid for $p > 4$, combined with the approximation and Fatou argument of Remark 9.2, gives $\mathbb{E}|X_{G_s}|^p \le c^p\mathbb{E}|\Pi_{\lambda/2} - \Pi'_{\lambda/2}|^p = \mathbb{E}|X_{(G_*)_s}|^p$. Adding up, $\mathbb{E}|S_{\mathbf{X}}|^p \le c^p\mathbb{E}|\Pi_\lambda - \lambda|^p$. For the reverse inequality, [7, Lemma 2.1] gives $\mathbf{X}^{(n)} \in \mathscr{X}_{p;A,B}$ with $S_{\mathbf{X}^{(n)}} \to c(\Pi_\lambda - \lambda)$ in distribution, and Fatou's lemma for weak convergence gives $\liminf_n \mathbb{E}|S_{\mathbf{X}^{(n)}}|^p \ge c^p\mathbb{E}|\Pi_\lambda - \lambda|^p$. □

Remark 9.4 (Numerical evidence for (9); not a proof). In Fourier form, (9) reads $\int e^{-a}(1-\cos b)\,t^{-p-1}dt \le \int e^{-a_*}(1-\cos b_*)\,t^{-p-1}dt$ with $a_* = \lambda(1-\cos ct)$, $b_* = \lambda(\sin ct - ct)$. We evaluated both sides with 30-digit quadrature and series-stabilized integrands for 180 random Lévy measures with 1–3 atoms, at $p \in \{4.2, 4.5, 4.8, 5.5\}$. The identity of Proposition 9.1(b) matched direct summation to 7 digits, and the ratio of the two sides never exceeded 1; equality holds only for single (or coincident) atoms, and the largest ratio for genuinely multi-atom G was 0.9954. In the subfamily $G = \lambda_+\delta_c + \lambda_-\delta_{-c}$, $\lambda_+ + \lambda_- = \lambda$, the symmetric part is fixed, so (9) there is *equivalent* to the restriction of the Poissonized conjecture $\mathbb{E}|\tilde{X}_G|^p \le c^p\mathbb{E}|\Pi_\lambda - \lambda|^p$ to this subfamily. (These G have nonzero

mean when $\lambda_+ \neq \lambda_-$; the centred laws $\tilde{X}_G$ enter the zero-mean problem through [7, Lemma 2.1].) Direct evaluation for $\lambda \in [0.1, 100]$ and $p \in \{4.2, 4.5, 4.8\}$ gives the maximum at $\lambda_- = 0$. Finally, for fixed G_s (180 random cases) the all-same-sign assignment maximized $\mathbb{E}|\tilde{X}_G|^p$ among all sign splittings. All scripts are in the accompanying repository (`numerics/split_45_stable.py` and others).

Remark 9.5 (Why second-variation arguments stop at $p = 5$; numerical). In the family $G_\theta = \frac{\lambda}{2}\big((1+\theta)\delta_1 + (1-\theta)\delta_{-1}\big)$, $\theta \in [-1, 1]$, put $Y_\theta = \tilde{X}_{G_\theta}$. Proposition 9.1(b) and the identity $\mathbb{E}Y_\theta^4 = \lambda + 3\lambda^2$ give

$$c_p' \frac{d^2}{d\theta^2}\mathbb{E}|Y_\theta|^p = \int_0^\infty e^{-\lambda(1-\cos t)}\lambda^2(t-\sin t)^2 \cos\big(\theta\lambda(t-\sin t)\big)\, t^{-p-1}dt.$$

For large λ the integrand is concentrated near $t \in 2\pi\mathbb{Z}$. The bump at $t = 0$ is positive and of order $\lambda^{(p-2)/2}$. The bumps at $t = 2\pi k$, $k \geq 1$, contribute $\approx \sqrt{2\pi}\lambda^{3/2}\sum_k (2\pi k)^{1-p}\cos(2\pi k\theta\lambda)$, of either sign. Hence for $4 < p < 5$ the map $\theta \mapsto \mathbb{E}|Y_\theta|^p$ is *not* convex when λ is large and $\theta\lambda \equiv \frac{1}{2} \pmod 1$. A 30-digit evaluation at $p = 4.5$, $\lambda = 2 \cdot 10^4$, $\theta = 10000.5/\lambda$ (so $\theta\lambda \equiv \frac{1}{2}$) gives the value -3769 (bump at 0: $+6808$; lattice: -10577); at integer $\theta\lambda$ the lattice part is $+12848$. For $p > 5$ the bump at 0 dominates: at $p = 5.5$, same λ, the bump is $2.01 \cdot 10^6$ and the lattice part is $-1.75 \cdot 10^3$ at $\theta\lambda \equiv \frac{1}{2}$ ($+1.91 \cdot 10^3$ at integer $\theta\lambda$). The asymptotic description above is a Laplace-method heuristic without error bounds; the non-convexity is established numerically at the computed point. Nevertheless the endpoint $\theta = \pm 1$ (Poisson) remains the maximizer in all cases computed. At first order near $\theta = 1$ the $t = 0$ bump, of order $\lambda^{(p-2)/2}$, dominates the lattice terms, of order $\lambda^{1/2}$ (heuristically). Thus, for $4 < p < 5$, $H \mapsto \mathbb{E}|Y_H|^p$ is not convex along this segment of Lévy measures, hence not convex on the constraint set; arguments relying on such convexity cannot work there. This is consistent with the sign change of the factor $(q-4)(q-5)$ in the proofs of [7, Props. 2.9 and 2.11], and points to the lattice structure of the Poisson characteristic function as the source of the difficulty.

Proposition 9.6 (Second-order optimality of the Poisson law in the rare-jump regime)**.** *For a finite Borel measure G on $\mathbb{R} \setminus \{0\}$ with $\int |x|^p dG < \infty$ put*

$$Q(G) := \tfrac{1}{2}\iint K(x,y)\, G(dx)G(dy), \qquad K(x,y) := |x+y|^p - |x|^p - |y|^p - p|x|^{p-2}xy - p|y|^{p-2}yx.$$

(a) *(Exact expansion.) Let $p \geq 3$ and let G_1 be such a measure. As $\varepsilon \downarrow 0$, $\mathbb{E}|\tilde{X}_{\varepsilon G_1}|^p = \varepsilon\int |x|^p dG_1 + \varepsilon^2 Q(G_1) + O(\varepsilon^3)$.*

(b) *(Representation.) Let $p > 4$ and let V, V' be i.i.d. with density $2(1-v)$ on $[0,1]$. Then $K(x,y) = \frac{p(p-1)(p-2)(p-3)}{4}\, x^2y^2\, \mathbb{E}|xV + yV'|^{p-4}$ for all x, y.*

(c) *(Optimality.) Let $4 < p \leq 5$, $\int x^2 dG = B > 0$, $\int |x|^p dG = A$, and let $c, \lambda > 0$ be defined by $c^p\lambda = A$, $c^2\lambda = B$. Then $Q(G) \leq Q(\lambda\delta_c)$, with equality if and only if $G \in \{\lambda\delta_c, \lambda\delta_{-c}\}$.*

(d) *(Consequence.) Let $4 < p \leq 5$ and let G_1 (with $\int x^2 dG_1 > 0$) not be a single atom. Then for all sufficiently small $\varepsilon > 0$, with $(A, B) = (\varepsilon\int |x|^p dG_1, \varepsilon\int x^2 dG_1)$, one has $\mathbb{E}|\tilde{X}_{\varepsilon G_1}|^p < c^p\mathbb{E}|\Pi_\lambda - \lambda|^p$.*

Proof. Write $f(x) = |x|^p$ and $q = G_1(\mathbb{R})$, $m = \int x\, dG_1$, ξ_k i.i.d. with law G_1/q, $N \sim \text{Poisson}(\varepsilon q)$, so $\tilde{X}_{\varepsilon G_1} \overset{d}{=} \sum_{k \leq N}\xi_k - \varepsilon m$.

(a) Condition on N. The event $N = 0$ contributes $e^{-\varepsilon q}f(-\varepsilon m) = O(\varepsilon^p) = O(\varepsilon^3)$ (here $p \geq 3$ is used). For $N = 1$, Taylor's formula with $f \in C^2$ and $\mathbb{E}|\xi|^p < \infty$ gives $\mathbb{E}f(\xi - \varepsilon m) = \mathbb{E}f(\xi) - \varepsilon m\mathbb{E}f'(\xi) + O(\varepsilon^2)$. For $N = 2$, $\mathbb{E}f(\xi_1 + \xi_2 - \varepsilon m) = \mathbb{E}f(\xi_1 + \xi_2) + O(\varepsilon)$. Finally $\sum_{k \geq 3}\mathbb{P}(N = k)\mathbb{E}|f(S_k - \varepsilon m)| \leq \sum_{k \geq 3}\mathbb{P}(N = k)C(1 + k^p) = O(\varepsilon^3)$. Collecting terms,

$$\mathbb{E}f(\tilde{X}_{\varepsilon G_1}) = \varepsilon q\mathbb{E}f(\xi) + \varepsilon^2\Big[\tfrac{q^2}{2}\mathbb{E}f(\xi_1 + \xi_2) - q^2\mathbb{E}f(\xi) - qm\mathbb{E}f'(\xi)\Big] + O(\varepsilon^3),$$

and the bracket equals $\frac{1}{2}\iint [f(x+y) - f(x) - f(y) - yf'(x) - xf'(y)]dG_1 dG_1 = Q(G_1)$.

(b) For $p>4$, $f\in C^4(\mathbb{R})$ with $f(0)=f'(0)=f''(0)=0$ and $f''''(x)=p(p-1)(p-2)(p-3)|x|^{p-4}$. Put $\phi(x,y)=f(x+y)-f(x)-f(y)-yf'(x)-xf'(y)$; then $\phi(x,0)=\phi(0,y)=0$ and $\partial_x\partial_y\phi=\psi(x,y):=f''(x+y)-f''(x)-f''(y)$, while $\psi(u,0)=\psi(0,w)=0$ and $\partial_u\partial_w\psi=f''''(u+w)$. Hence $\phi(x,y)=\int_0^x\int_0^y\int_0^u\int_0^w f''''(a+b)\,db\,da\,dw\,du$ (oriented integrals). The substitution $u=\sigma x$, $w=\tau y$, $a=\sigma' u$, $b=\tau' w$ gives $\phi(x,y)=x^2y^2\int_{[0,1]^4}\sigma\tau f''''(\sigma\sigma' x+\tau\tau' y)\,d\sigma\,d\sigma'\,d\tau\,d\tau'$. Finally $\int_0^1\int_0^1\sigma g(\sigma\sigma')d\sigma'd\sigma=\int_0^1(1-v)g(v)dv=\frac12\mathbb{E}g(V)$ for bounded measurable g, applied in (σ,σ') and in (τ,τ').

(c) Let $H=x^2G$, a measure of mass B, and let Y_1,Y_2 be i.i.d. with law H/B, independent of (V,V'); then $\mathbb{E}|Y|^{p-2}=A/B=c^{p-2}$. By (b), with $r:=p-4\in(0,1]$ and $D_p:=p(p-1)(p-2)(p-3)/8$,

$$\begin{aligned}Q(G)=D_pB^2\,\mathbb{E}|Y_1V+Y_2V'|^r&\le D_pB^2\,\mathbb{E}\big(|Y_1|V+|Y_2|V'\big)^r\\&\le D_pB^2\,(\mathbb{E}|Y|)^r\,\mathbb{E}(V+V')^r\le D_pB^2c^r\mathbb{E}(V+V')^r.\end{aligned}$$

Here we used, in turn: the triangle inequality and $V,V'\ge0$; Jensen's inequality conditionally on (V,V') for the concave function $(R_1,R_2)\mapsto(R_1V+R_2V')^r$ on $[0,\infty)^2$ ($0<r\le1$; note $\mathbb{E}|Y|=\int|x|^3dG/B<\infty$); and $\mathbb{E}|Y|\le(\mathbb{E}|Y|^{p-2})^{1/(p-2)}=c$ (Lyapunov, $p-2>1$). The right-hand side is $Q(\lambda\delta_c)$, since then $Y\equiv c$. If equality holds, Lyapunov's inequality forces $|Y|=c$ a.s., and equality in the triangle inequality ($V,V'>0$ a.s.) forces Y_1,Y_2 to have a.s. the same sign; as Y_1,Y_2 are i.i.d., $\mathbb{P}(Y>0)^2+\mathbb{P}(Y<0)^2=1$, so Y has a constant sign. Thus $G=\lambda\delta_c$ or $\lambda\delta_{-c}$.

(d) The Poisson competitor of εG_1 is $\varepsilon G_{1,*}$ with $G_{1,*}=\lambda'\delta_c$ and fixed c; so (a) applied to G_1 and to $G_{1,*}$ gives $\mathbb{E}|\tilde X_{\varepsilon G_1}|^p-\mathbb{E}|\tilde X_{\varepsilon G_{1,*}}|^p=\varepsilon^2(Q(G_1)-Q(G_{1,*}))+O(\varepsilon^3)$, and $Q(G_1)<Q(G_{1,*})$ by (c). □

Proof of Theorem H. By homogeneity ($X_i\mapsto X_i/c$) it suffices to take $c=1$, so $A=B=\lambda$; then $B^{p/2}=\lambda^{p/2}$. The lower bound is [7, Lemma 2.1] with Fatou's lemma. For the upper bound let $\mathbf{X}\in\mathscr{X}_{p;\lambda,\lambda}$ and $G=\sum_i G_{X_i}$; by Corollary 2.5, $\mathbb{E}|S_{\mathbf{X}}|^p\le\mathbb{E}|X_G|^p$, where $\int x\,dG=0$ and $\int x^2dG=\int|x|^pdG=\lambda$. For any finite G on $\mathbb{R}\setminus\{0\}$ with $\int x^2dG=\int|x|^pdG=\lambda$ (not necessarily centred), let $\psi(t)=\int(e^{itx}-1-itx)G(dx)=-a(t)+ib(t)$, the exponent of the centred law $\tilde X_G=X_G-\int x\,dG$, and $g(t)=\lambda t^2/2$. Being a shift of X_G, $\tilde X_G$ has a finite p-th moment; the proof of Lemma 2.3 gives $\mathbb{E}|X_G|^p<\infty$ without using $\int x\,dG=0$. As in the proof of Proposition G (with $\mathbb{E}\tilde X_G=0$, $\mathbb{E}\tilde X_G^2=\lambda$, $\mathbb{E}\tilde X_G^4=\int x^4dG+3\lambda^2$ and $\int x^4dG\le\int(x^2+|x|^p)dG<\infty$),

$$c_p'\big(\mathbb{E}|\tilde X_G|^p-\lambda\big)=\int_0^\infty\Big(1-a-e^{-a}\cos b+\frac{g^2}{2}\Big)t^{-p-1}dt=\int_0^\infty\Big(\frac{g^2-\operatorname{Re}\psi^2}{2}-\operatorname{Re}\rho(\psi)\Big)t^{-p-1}dt,$$

where $\rho(z)=e^z-1-z-z^2/2$; here $1-a-e^{-a}\cos b=-\operatorname{Re}(e^\psi-1-\psi)$. Since $|e^{iy}-1-iy|\le y^2/2$, we have $|\psi(t)|\le g(t)$. For $\operatorname{Re}z\le0$, $|\rho(z)|\le|z|^3/6$ and $|\rho(z)|\le|e^z-1-z|+|z|^2/2\le|z|^2$. Hence

$$\int_0^\infty|\rho(\psi)|t^{-p-1}dt\le\int_0^\infty\min\Big(\frac{g^3}{6},g^2\Big)t^{-p-1}dt=\eta_p\lambda^{p/2},$$

by splitting at $t^2=12/\lambda$. In particular the two parts of the integrand are separately integrable (near $t=0$, $g^2-\operatorname{Re}\psi^2=O(t^6)$; at ∞, $|g^2-\operatorname{Re}\psi^2|\le2g^2$). The quadratic part equals $c_p'Q(G)$: apply the display to εG, in which ψ, g scale by ε; the ρ-term is $O(\varepsilon^{p/2})=o(\varepsilon^2)$; comparing with Proposition 9.6(a) identifies the ε^2-coefficients. We record the outcome, which is used again in the proofs of Theorems I–K: for every finite G on $\mathbb{R}\setminus\{0\}$ with $\int x^2dG=\int|x|^pdG=\lambda$, centred or not,

$$\mathbb{E}|\tilde X_G|^p=\lambda+Q(G)-\frac{1}{c_p'}\int_0^\infty\operatorname{Re}\rho(\psi_G(t))\,t^{-p-1}dt,\qquad\psi_G(t)=\int(e^{itx}-1-itx)G(dx).\tag{10}$$

By Proposition 9.6(c), $Q(G) \le Q(\lambda\delta_1) = (2^{p-1}-p-1)\lambda^2$. Therefore, for our $G = \sum_i G_{X_i}$ (for which $\tilde{X}_G = X_G$),

$$\mathbb{E}|X_G|^p \le \lambda + q_p\lambda^2 + \eta_p\lambda^{p/2}/c_p',$$

and taking the supremum over $\mathbf{X}$ gives the upper bound for $\mathcal{E}_{p;\lambda,\lambda}$. For $G_* = \lambda\delta_1$ (the centred Poisson law, $\tilde{X}_{G_*} = \Pi_\lambda - \lambda$) one has $Q(G_*) = q_p\lambda^2$, so the display gives $|\mathbb{E}|\Pi_\lambda - \lambda|^p - \lambda - q_p\lambda^2| \le \eta_p\lambda^{p/2}/c_p'$. Scaling back ($c^p\lambda = A$, $c^p\lambda^2 = A\lambda$, $c^p\lambda^{p/2} = B^{p/2}$) gives the displays of the theorem, and the remaining assertions follow by combining them. Here $c_p' = \int_0^\infty k_4(s)s^{-p-1}ds$, and the closed form is the classical integral $\int_0^\infty(\cos s - 1 + \frac{s^2}{2} - \frac{s^4}{24})s^{-p-1}ds = \Gamma(-p)\cos\frac{\pi p}{2}$ for $4 < p < 6$. It follows from $\int_0^\infty s^{w-1}\sin s\,ds = \Gamma(w)\sin\frac{\pi w}{2}$ $(-1 < \operatorname{Re} w < 1)$ by repeated integration by parts. It is used only to evaluate constants, namely c_p' in Theorem H and K_p in Theorem J. An alternative to the scaling step is the pointwise identity $c_p'K(x,y) = \int_0^\infty[\frac{1}{4}t^4x^2y^2 - (1-\cos tx)(1-\cos ty) + (\sin tx - tx)(\sin ty - ty)]t^{-p-1}dt$. □

The rare-jump regime: proof of Theorem I. By homogeneity we take $c = 1$, so $A = B = \lambda$. Let $\mathscr{G}_\lambda$ be the set of finite Borel measures G on $\mathbb{R}\setminus\{0\}$ with $\int x^2dG = \int|x|^pdG = \lambda$. For $G \in \mathscr{G}_\lambda$ let $Y = Y_G$ be a random variable with law $x^2G(dx)/\lambda$; then $\mathbb{E}|Y|^{p-2} = 1$ and $\mathbb{E}|Y| \le 1$. Put $G_* := \lambda\delta_1$, $r := p-4 \in (0,1)$, $D_p := p(p-1)(p-2)(p-3)/8$, $e_r := \mathbb{E}(V+V')^r$ with V, V' as in Proposition 9.6, and

$$\omega_0(s) := \frac{p-3}{p-2} + \frac{s^{p-2}}{p-2} - s \quad (s \ge 0), \qquad \hat{\omega}(y) := \omega_0(|y|) + \mathbf{1}_{\{y<0\}}\big(1+|y|^{p-2}\big) \quad (y \in \mathbb{R}).$$

By the weighted AM–GM inequality, $\omega_0 \ge 0$ with equality only at $s = 1$; hence $\hat{\omega} \ge 0$ with equality only at $y = 1$. Since $\omega_0(1) = \omega_0'(1) = 0$ and $\omega_0''(s) = (p-3)s^{p-4} \ge \frac{1}{2}$ on $[\frac{1}{2}, 2]$, we have $\omega_0(s) \ge \frac{1}{4}(s-1)^2$ there. Moreover $C_\omega := \sup\{(1+s^{p-2})/\omega_0(s) :\ s \in [0,\frac{1}{2}) \cup (2,\infty)\} < \infty$, because ω_0 is continuous and positive on $[0,\frac{1}{2}] \cup [2,\infty)$ and $(1+s^{p-2})/\omega_0(s) \to p-2$ as $s \to \infty$. Finally $\mathbb{E}\omega_0(|Y|) = 1 - \mathbb{E}|Y|$, since $\mathbb{E}|Y|^{p-2} = 1$.

Lemma 9.7 (Stability of Proposition 9.6(c))**.** *There is $c_1 = c_1(p) > 0$ such that for all $\lambda > 0$ and all $G \in \mathscr{G}_\lambda$ with $\mathbb{P}(Y \in [\frac{1}{2}, 2]) \ge \mathbb{P}(Y \in [-2, -\frac{1}{2}])$,*

$$Q(G_*) - Q(G) \ \ge\ c_1\lambda^2\,\mathbb{E}\hat{\omega}(Y).$$

Proof. Let Y_1, Y_2 be independent copies of Y, independent of (V, V'), and let $u := \mathbb{E}|Y|$. As in the proof of Proposition 9.6(c), $Q(G) = D_p\lambda^2\mathbb{E}|Y_1V + Y_2V'|^r$, $Q(G_*) = D_p\lambda^2e_r$, and $\mathbb{E}(|Y_1|V + |Y_2|V')^r \le u^re_r$ by Jensen's inequality, applied conditionally on (V, V'). Hence

$$Q(G_*) - Q(G) \ge D_p\lambda^2\big[(1-u^r)e_r + \Sigma\big], \qquad \Sigma := \mathbb{E}\big[(|Y_1|V + |Y_2|V')^r - |Y_1V + Y_2V'|^r\big] \ge 0.$$

Since $u \mapsto 1-u^r$ is convex on $[0,1]$, $1-u^r \ge r(1-u) = r\mathbb{E}\omega_0(|Y|)$. For $x, y \in [0,2]$ the mean value theorem gives $(x+y)^r - |x-y|^r \ge r4^{r-1}\big((x+y)-|x-y|\big) = 2r4^{r-1}\min(x,y)$. Applied with $x = aV$, $y = bV'$, $a, b \in [\frac{1}{2}, 2]$, and combined with $\mathbb{E}\min(V,V') = \int_0^1\mathbb{P}(V > s)^2ds = \int_0^1(1-s)^4ds = \frac{1}{5}$, this gives $\mathbb{E}[(aV+bV')^r - |aV-bV'|^r] \ge g_* := r4^{r-1}/5$. The integrand of Σ vanishes unless $Y_1Y_2 < 0$, and the joint law and the integrand are invariant under exchanging (Y_1, V) with (Y_2, V'). Hence, with $\pi_\pm := \mathbb{P}(\pm Y \in [\frac{1}{2}, 2])$, conditioning on (Y_1, Y_2) gives $\Sigma \ge 2g_*\pi_+\pi_-$. Next, $1 + |y|^{p-2} \le 9$ for $|y| \le 2$, and $1 \le 1 + |y|^{p-2} \le C_\omega\omega_0(|y|)$ for $|y| \notin [\frac{1}{2}, 2]$; therefore

$$\mathbb{E}\hat{\omega}(Y) \le (1+C_\omega)\mathbb{E}\omega_0(|Y|) + 9\pi_-, \qquad \mathbb{P}\big(|Y| \notin [\tfrac{1}{2}, 2]\big) \le C_\omega\mathbb{E}\omega_0(|Y|).$$

If $\mathbb{E}\omega_0(|Y|) \ge 1/(4C_\omega)$, then $9\pi_- \le 36C_\omega\mathbb{E}\omega_0(|Y|)$, and $Q(G_*) - Q(G) \ge D_p\lambda^2re_r\mathbb{E}\omega_0(|Y|) \ge \frac{D_pre_r}{1+37C_\omega}\lambda^2\mathbb{E}\hat{\omega}(Y)$. Otherwise $\pi_+ + \pi_- > \frac{3}{4}$, hence $\pi_+ > \frac{3}{8}$ by hypothesis, so $\Sigma \ge \frac{3}{4}g_*\pi_-$ and

$$\mathbb{E}\hat{\omega}(Y) \le M_1\Big(re_r\mathbb{E}\omega_0(|Y|) + \tfrac{3}{4}g_*\pi_-\Big) \le M_1\,\frac{Q(G_*) - Q(G)}{D_p\lambda^2}, \qquad M_1 := \max\Big\{\frac{1+C_\omega}{re_r}, \frac{12}{g_*}\Big\}.$$

Both cases give the claim with $c_1 := D_p \min\{re_r/(1+37C_\omega),\, 1/M_1\}$. □

Lemma 9.8 (A constraint-adapted bound for the Lévy exponent)**.** *For $t>0$ and $y\in\mathbb{R}$ let $h_t(y) := (e^{ity}-1-ity)/y^2$, $h_t(0) := -t^2/2$, and $e_t(y) := h_t(y) - h_t(1) - \frac{h_t'(1)}{p-2}\big(|y|^{p-2}-1\big)$. Then $|e_t(y)| \le C_M t^2 \hat\omega(y)$ for all $t>0$ and $y\in\mathbb{R}$, where $C_M := \max\{968,\, 5\max(1, C_\omega)\}$. Consequently, for $G\in\mathscr{G}_\lambda$ and $\psi_G(t) := \int (e^{itx}-1-itx)G(dx)$,*

$$|\psi_G(t) - \psi_{G_*}(t)| \le C_M\, t^2\, \lambda\, \mathbb{E}\hat\omega(Y) \qquad (t>0).$$

Proof. From $e^{iz}-1-iz = -z^2\int_0^1 (1-u)e^{iuz}du$ we get $h_t(y) = -t^2\int_0^1(1-u)e^{ituy}du$. So h_t is smooth, with $|h_t|\le t^2/2$, $|h_t'|\le t^3/6$ and $|h_t''|\le t^4/12$. Also, with $\phi(y) := e^{ity}-1-ity$, we have $h_t'(1) = \phi'(1) - 2\phi(1)$ and $h_t'' = \phi'' y^{-2} - 4\phi' y^{-3} + 6\phi y^{-4}$ for $y\neq 0$, where $|\phi''| = t^2$, $|\phi'|\le 2t$ and $|\phi(y)|\le 2+t|y|$. Hence $|h_t'(1)|\le 4t+4$ and, on $[\frac12, 2]$, $|h_t''|\le 4t^2+256t+192$. Using the first set of bounds for $t\le 1$ and the second for $t>1$, we get $|h_t'(1)|\le 8t^2$, and $|h_t''|\le 452t^2$ on $[\frac12,2]$, for all $t>0$.

On $[\frac12,2]$ we have $e_t(1) = e_t'(1) = 0$ and $e_t''(y) = h_t''(y) - (p-3)y^{p-4}h_t'(1)$, with $0<(p-3)y^{p-4}\le 4$. Hence $|e_t''|\le 452t^2 + 32t^2 = 484t^2$ there, and Taylor's formula with integral remainder gives $|e_t(y)|\le 242t^2(y-1)^2 \le 968t^2\omega_0(y) = 968t^2\hat\omega(y)$. Off $[\frac12,2]$, $|e_t(y)|\le |h_t(y)| + |h_t(1)| + \frac{|h_t'(1)|}{p-2}\big(1+|y|^{p-2}\big) \le t^2 + 4t^2\big(1+|y|^{p-2}\big) \le 5t^2\big(1+|y|^{p-2}\big)$, and $1+|y|^{p-2}\le \max(1,C_\omega)\hat\omega(y)$ there (for $y<0$ by the definition of $\hat\omega$, for $y\ge 0$ by that of C_ω).

For the consequence, let $H := x^2 G$, so that $H(\mathbb{R}) = \int |x|^{p-2}dH = \lambda$, $\psi_G(t) = \int h_t\, dH$ and $\psi_{G_*}(t) = \lambda h_t(1)$. The two constraints give $\psi_G(t) - \psi_{G_*}(t) = \int e_t\, dH$, and $\int |e_t|\, dH \le C_M t^2 \int \hat\omega\, dH = C_M t^2 \lambda\, \mathbb{E}\hat\omega(Y)$. □

Proof of Theorem I. Let $G\in\mathscr{G}_\lambda$, $\Phi(G) := \mathbb{E}|\tilde X_G|^p$ and $\Omega := \mathbb{E}\hat\omega(Y_G)$. Replacing G by its reflection $G(-\,\cdot\,)$ changes neither $\Phi(G)$ nor $Q(G)$, and replaces Y by $-Y$; so we may assume $\mathbb{P}(Y\in[\frac12,2]) \ge \mathbb{P}(Y\in[-2,-\frac12])$. By (10), $\Phi(G) = \lambda + Q(G) - \frac{1}{c_p'}\int_0^\infty \operatorname{Re}\rho(\psi_G)\, t^{-p-1}dt$ with $\rho(z) = e^z - 1 - z - z^2/2$, and likewise for G_*. Hence

$$\Phi(G) - \Phi(G_*) = -\big(Q(G_*) - Q(G)\big) - \frac{1}{c_p'}\int_0^\infty \operatorname{Re}\big(\rho(\psi_G) - \rho(\psi_{G_*})\big)t^{-p-1}dt.$$

For $\operatorname{Re} z\le 0$, $\rho'(z) = e^z - 1 - z$ satisfies $|\rho'(z)|\le \min(|z|^2/2, 2|z|)$. Every point z of the segment $[\psi_{G_*}(t), \psi_G(t)]$ has $\operatorname{Re} z\le 0$ and $|z|\le g(t) = \lambda t^2/2$. So, by Lemma 9.8,

$$\big|\rho(\psi_G(t)) - \rho(\psi_{G_*}(t))\big| \le |\psi_G(t) - \psi_{G_*}(t)| \min\Big(\frac{g^2}{2}, 2g\Big) \le C_M \lambda\Omega\, t^2 \min\Big(\frac{\lambda^2 t^4}{8}, \lambda t^2\Big).$$

The substitution $t = s/\sqrt{\lambda}$ gives

$$\int_0^\infty t^2 \min\Big(\frac{\lambda^2 t^4}{8}, \lambda t^2\Big)t^{-p-1}dt = j_p \lambda^{(p-2)/2}, \qquad \text{where}$$

$$j_p := \int_0^\infty s^2 \min\Big(\frac{s^4}{8}, s^2\Big)s^{-p-1}ds = \frac{8^{(6-p)/2}}{8(6-p)} + \frac{8^{(4-p)/2}}{p-4} < \infty.$$

Together with Lemma 9.7 this gives

$$\Phi(G) - \Phi(G_*) \le -c_1\lambda^2\Omega + \frac{C_M j_p}{c_p'}\lambda^{p/2}\Omega = \lambda^2\Omega\Big(\frac{C_M j_p}{c_p'}\lambda^{(p-4)/2} - c_1\Big).$$

Let $\lambda_0(p) := \big(c_1 c_p'/(2C_M j_p)\big)^{2/(p-4)}$. For $\lambda\le\lambda_0(p)$ the bracket is at most $-c_1/2$, so $\Phi(G)\le\Phi(G_*)$. Equality forces $\Omega = 0$, i.e. $Y = 1$ a.s., i.e. $G = G_*$; before the reflection this means $G\in\{\lambda\delta_1, \lambda\delta_{-1}\}$. Since $\Phi(G_*) = \mathbb{E}|\Pi_\lambda - \lambda|^p$, this proves the second assertion (for $c=1$). For the first, let $\mathbf{X}\in\mathscr{X}_{p;\lambda,\lambda}$ and $G := \sum_i G_{X_i}\in\mathscr{G}_\lambda$. This G is centred, so $\tilde X_G = X_G$, and Corollary 2.5 gives $\mathbb{E}|S_{\mathbf{X}}|^p \le \Phi(G) \le$

$\mathbb{E}|\Pi_\lambda - \lambda|^p$. The reverse inequality for the supremum is [7, Lemma 2.1] with Fatou's lemma. Scaling back gives the theorem. □

Remark 9.9 (Size of λ_0; numerical). Take g_* as above and $e_r = q_p/D_p$. The supremum defining C_ω is the limit as $s \downarrow 2$, as a grid evaluation confirms, so $C_\omega = (1 + 2^{p-2})/\omega_0(2) \approx 8.82, 7.72, 7.05$ at $p = 4.2, 4.5, 4.8$. With these values, the definition $\lambda_0 = (c_1 c_p'/(2C_M j_p))^{2/(p-4)}$ gives $\lambda_0 \approx 2 \cdot 10^{-74}$, $4 \cdot 10^{-28}$, $4 \cdot 10^{-17}$. The loss comes from the exponent $2/(p-4)$ and from crude constants. The best constant in Lemma 9.7 is at most about $0.13, 0.45, 0.96$, versus the proved $c_1 \approx 0.0024, 0.010, 0.025$; these upper bounds are attained by an all-negative two-point Y just outside $[-2, -\frac{1}{2}]$. The best constant in Lemma 9.8 is about $1.5, 1.3, 1.1$, versus $C_M = 968$. With these values the same argument would give $\lambda_0 \approx 5 \cdot 10^{-29}$, $5 \cdot 10^{-10}$, $8 \cdot 10^{-6}$ (`numerics/check_small_lambda.py`). As $p \downarrow 4$, c_1 is of order $p - 4$, so λ_0 tends to 0 super-exponentially. This is intrinsic to the method: at $p = 4$ the constraints fix $\mathbb{E}S^4$ and the deficit of Q disappears. A substantially larger λ_0 would need a remainder bound better than $O(\lambda^{p/2}\Omega)$, for instance one separating the small jumps of G.

Remark 9.10. (i) The argument in (c) uses $p \le 5$ only through the concavity of $t \mapsto t^{p-4}$; $p = 5$ is a threshold of this method only. For $p \ge 5$ the conclusion $Q(G) \le Q(\lambda\delta_c)$ also holds: by [7, Lemma 2.1] and Fatou, $\mathbb{E}|\tilde{X}_{\varepsilon G}|^p \le \mathcal{E}_{p;\varepsilon A,\varepsilon B}$, which equals the Poisson value by [7, Thm. 1.3]; expanding both sides by (a) gives the claim. (A sufficient pointwise condition, $\vartheta(1,s) \le \vartheta(1,1)((1+|s|^{p-2})/2)^{(p-4)/(p-2)}$ with $\vartheta = K/(x^2y^2)$, also holds on a numerical grid for $p \in [4.05, 5.9]$.) (ii) Proposition 9.6(d) is a statement along rays: the $O(\varepsilon^3)$ term depends on G_1. The uniform statements are Theorem H, whose remainder $O(B^{p/2})$ is uniform but only of relative size $O(\lambda^{(p-4)/2})$ compared with the second-order term, and Theorem I. In the latter, Lemma 9.7 is the quantitative form of (c), and Lemma 9.8 makes the remainder vanish to the same order as the deficit, $\mathbb{E}\hat{\omega}(Y)$, near $\lambda\delta_1$. Without the subtraction of $h_t(1) + \frac{h_t'(1)}{p-2}(|y|^{p-2} - 1)$ the remainder would only be of the order of the distance of Y from 1, whereas the deficit is of the order of its square. (iii) The representation (b) extends to $3 < p < 4$ (then $f'''' \propto |x|^{p-4}$ is locally integrable and f''' is absolutely continuous, which suffices for the same integrations). There $r = p - 4 < 0$, and small jumps (Gaussian-like parts) make Q large: e.g. at $p = 3.5$, $G = \delta_1 + n^2\delta_{1/n}$ gives $Q(G)/Q(\lambda\delta_c) = 1.43, 2.88, 7.21$ for $n = 10, 100, 1000$. This is consistent with Theorem A.

The Gaussian regime: proof of Theorem J. Again $c = 1$, $G \in \mathscr{G}_\lambda$, $g(t) = \lambda t^2/2$, and $\psi_G(t) = \int (e^{itx} - 1 - itx)G(dx) = -a(t) + ib(t)$. Put

$$\varpi(u) := e^{iu} - 1 - iu + \tfrac{u^2}{2}, \qquad k_2(u) := \operatorname{Re}\varpi(u) = \cos u - 1 + \tfrac{u^2}{2} \ge 0,$$
$$\psi_r(t) := \int \varpi(tx)\, G(dx) = \psi_G(t) + g(t),$$

so that $\operatorname{Re}\psi_r = g - a \in [0, g]$ and $\operatorname{Im}\psi_r = b$. From $|\varpi(u)| \le |u|^3/6$, $k_2(u) \le u^4/24$ and the interpolation inequalities $\int |x|^3 dG \le \lambda$, $\int x^4 dG \le \lambda$ (weighted AM–GM: $|x|^q \le \theta x^2 + (1-\theta)|x|^p$ with $q = 2\theta + p(1-\theta)$), we get

$$|\psi_r(t)| \le \lambda t^3/6, \qquad 0 \le \operatorname{Re}\psi_r(t) \le \lambda t^4/24, \qquad |\psi_r(t)| \le |\psi_G(t)| + g(t) \le 2g(t). \tag{11}$$

Moreover $a(t) \ge \frac{11}{16} g(t)$ for $0 < t \le \frac{1}{2}$. Indeed, $1 - \cos u \ge \frac{11}{24}u^2$ for $|u| \le 1$, and $\int_{|x|>1/t} x^2 dG \le t^{p-2}\int |x|^p dG \le \lambda/4$, so $a(t) \ge \frac{11}{24}t^2 \cdot \frac{3}{4}\lambda$.

Lemma 9.11 (A second decomposition)**.** *For $G \in \mathscr{G}_\lambda$ and $\Phi(G) = \mathbb{E}|\tilde{X}_G|^p$,*

$$c_p'\big(\Phi(G) - \lambda - m_p\lambda^{p/2}\big) = L(G) + N(G), \qquad \textit{where}$$
$$L(G) := \int_0^\infty \operatorname{Re}\psi_r\,(1 - e^{-g})\, t^{-p-1}dt, \qquad N(G) := -\int_0^\infty e^{-g}\,\operatorname{Re}\big(e^{\psi_r} - 1 - \psi_r\big)\, t^{-p-1}dt,$$

with absolutely convergent integrals.

Proof. The display in the proof of Theorem H gives $c_p'(\Phi(G)-\lambda) = \int_0^\infty (1-a-e^{-a}\cos b+\frac{g^2}{2})t^{-p-1}dt$, and the representation of $|x|^p$ by k_4 applied to $\lambda^{1/2}Z$ gives $c_p' m_p \lambda^{p/2} = \int_0^\infty (1-g-e^{-g}+\frac{g^2}{2})t^{-p-1}dt$. Both integrals converge absolutely. The difference of the integrands is $g-a-e^{-a}\cos b+e^{-g} = \operatorname{Re}\psi_r(1-e^{-g}) - e^{-g}\operatorname{Re}(e^{\psi_r}-1-\psi_r)$, because $e^{-a}\cos b = \operatorname{Re} e^{\psi_G} = e^{-g}\operatorname{Re} e^{\psi_r}$. The first term is nonnegative and, by (11), at most $\min(\lambda t^4/24, g)\min(g,1)$, which is integrable against t^{-p-1} for $4<p<6$. Hence the second term is integrable as well. □

Lemma 9.12 (The linear part is maximized by the Poisson law)**.** *For $\mu>0$ let $\Psi(\mu) := \int_0^\infty k_2(u)(1-e^{-\mu u^2/2})u^{-p-1}du$, and for $s>0$ let $\varphi_\lambda(s) := s\Psi(\lambda s^{-2/(p-2)})$. Then φ_λ is strictly concave on $(0,\infty)$, and $L(G) = \lambda\mathbb{E}\varphi_\lambda(|Y_G|^{p-2})$. Consequently $L(G)\le\lambda\varphi_\lambda(1) = L(G_*)$ for every $G\in\mathscr{G}_\lambda$, with equality if and only if $|Y_G|=1$ a.s.*

Proof. Ψ is finite, since $k_2(u)(1-e^{-\mu u^2/2}) \le \min(\mu u^6/48, u^2/2)$. Differentiation under the integral sign, justified by domination on $\mu\ge\mu_0>0$, gives $\Psi'(\mu) = \frac12\int k_2(u)u^{1-p}e^{-\mu u^2/2}du > 0$ and $\Psi''(\mu) = -\frac14\int k_2(u)u^{3-p}e^{-\mu u^2/2}du < 0$. With $\beta := 2/(p-2)\in(0,1)$ and $\mu = \lambda s^{-\beta}$, so that $d\mu/ds = -\beta\mu/s$, we get $\varphi_\lambda'(s) = \Psi(\mu)-\beta\mu\Psi'(\mu)$ and $\varphi_\lambda''(s) = -\frac{\beta\mu}{s}\big((1-\beta)\Psi'(\mu)-\beta\mu\Psi''(\mu)\big) < 0$. By Tonelli and the substitution $u = t|x|$, $L(G) = \int\kappa_\lambda\,dG$ with $\kappa_\lambda(x) = \int_0^\infty k_2(tx)(1-e^{-\lambda t^2/2})t^{-p-1}dt = |x|^p\Psi(\lambda/x^2)$, i.e. $\kappa_\lambda(x) = x^2\varphi_\lambda(|x|^{p-2})$. Hence $L(G) = \int\varphi_\lambda(|x|^{p-2})\,x^2G(dx) = \lambda\mathbb{E}\varphi_\lambda(|Y|^{p-2})$. Since $\mathbb{E}|Y|^{p-2}=1$ and $\varphi_\lambda\ge0$ is strictly concave, Jensen's inequality gives the claim and its equality case. □

Lemma 9.13 (The nonlinear part)**.** *With $\rho(z) = e^z-1-z-z^2/2$ as before, put*

$$N_s := \tfrac12\int_0^\infty e^{-g}(\operatorname{Im}\psi_r)^2t^{-p-1}dt, \qquad N_c := \tfrac12\int_0^\infty e^{-g}(\operatorname{Re}\psi_r)^2t^{-p-1}dt,$$
$$N_3 := \int_0^\infty e^{-g}\operatorname{Re}\rho(\psi_r)\,t^{-p-1}dt,$$

and $N_s^+ := \frac12\int_0^\infty e^{-g}(\lambda t^3/6)^2t^{-p-1}dt = n_1\lambda^{(p-2)/2}$ with $n_1 := 2^{(6-p)/2}\Gamma(\frac{6-p}{2})/144$. Then $N = N_s - N_c - N_3$ with absolutely convergent integrals. Moreover there is $C_N = C_N(p)$ such that for $\lambda\ge1$ and all $G\in\mathscr{G}_\lambda$,

$$0\le N_s(G)\le N_s^+, \qquad N_s(G_*)\ge N_s^+ - C_N\lambda^{(p-4)/2},$$
$$0\le N_c(G)\le C_N\lambda^{(p-4)/2}, \qquad |N_3(G)|\le C_N\lambda^{(p-4)/2}.$$

In particular $N(G)\le N_s^+ + C_N\lambda^{(p-4)/2}$ and $N(G_*)\ge N_s^+ - 3C_N\lambda^{(p-4)/2}$.

Proof. The identity $N = N_s - N_c - N_3$ follows from $\operatorname{Re}(z^2) = (\operatorname{Re}z)^2 - (\operatorname{Im}z)^2$. The integrals converge absolutely by the bounds below. By (11), $0\le N_s(G)\le N_s^+$, and for every $G\in\mathscr{G}_\lambda$, $0\le N_c(G)\le\frac12\int e^{-g}(\lambda t^4/24)^2t^{-p-1}dt = O(\lambda^{(p-4)/2})$. For G_*, $\operatorname{Im}\psi_r = -\lambda(t-\sin t)$ and $0\le(t^3/6)^2-(t-\sin t)^2\le t^8/360$, so $N_s^+ - N_s(G_*)\le\frac{\lambda^2}{720}\int e^{-g}t^{7-p}dt = O(\lambda^{(p-4)/2})$. For N_3 on $t\le\frac12$, write $\rho(z) = z^3/6+\rho_4(z)$ with $|\rho_4(z)|\le\frac{|z|^4}{24}e^{\max(\operatorname{Re}z,0)}$. With $\xi := \operatorname{Re}\psi_r\in[0,\lambda t^4/24]$ and $|\eta| := |\operatorname{Im}\psi_r|\le\lambda t^3/6$, the cubic term has $|\operatorname{Re}(\psi_r^3)| = |\xi^3-3\xi\eta^2|\le\frac{\lambda^3t^{12}}{13824}+\frac{\lambda^3t^{10}}{288}$: its dominant part $(i\eta)^3$ is purely imaginary. For the quartic term, $e^{-g}e^{\operatorname{Re}\psi_r} = e^{-a}\le e^{-11g/16}$ gives $e^{-g}|\rho_4(\psi_r)|\le e^{-11g/16}(\lambda t^3/6)^4/24$. On $t>\frac12$ we use $e^{-g}|\rho(\psi_r)|\le e^{-a}+e^{-g}(1+2g+2g^2)\le3$. Hence, uniformly in $G\in\mathscr{G}_\lambda$,

$$|N_3(G)|\le\int_0^\infty\Big[\frac{e^{-g}}{6}\Big(\frac{\lambda^3t^{12}}{13824}+\frac{\lambda^3t^{10}}{288}\Big)+e^{-11g/16}\frac{\lambda^4t^{12}}{31104}\Big]t^{-p-1}dt+3\int_{1/2}^\infty t^{-p-1}dt = O(\lambda^{(p-4)/2})+\frac{3\cdot2^p}{p}.$$

All $O(\cdot)$ terms are explicit multiples of the indicated powers (or of smaller ones), by $\int_0^\infty e^{-\alpha t^2}t^k dt = \frac12\alpha^{-(k+1)/2}\Gamma(\frac{k+1}{2})$. Collecting the terms and using $\lambda\ge1$ proves the lemma. □

Proof of Theorem J. Take $c = 1$. Let $G \in \mathscr{G}_\lambda$. By Lemmas 9.11–9.13, $c_p'(\Phi(G) - \Phi(G_*)) = (L(G)-L(G_*))+(N(G)-N(G_*)) \le 4C_N\lambda^{(p-4)/2}$. As in the proof of Theorem I, Poissonization and [7, Lemma 2.1] turn this into $\mathbb{E}|\Pi_\lambda - \lambda|^p \le \mathcal{E}_{p;\lambda,\lambda} \le \mathbb{E}|\Pi_\lambda - \lambda|^p + 4C_N\lambda^{(p-4)/2}/c_p'$. For the expansion of $\Phi(G_*)$, write $k_2(u) = u^4/24 - k_4(u)$. With $\int_0^\infty u^{3-p}(1 - e^{-\mu u^2/2})du = -\frac{1}{2}\Gamma(\frac{4-p}{2})(\frac{\mu}{2})^{(p-4)/2}$ (valid for $4 < p < 6$), this gives $\Psi(\mu) = l_1\mu^{(p-4)/2} - c_p' + \int_0^\infty k_4(u)e^{-\mu u^2/2}u^{-p-1}du$ with $l_1 := -2^{(4-p)/2}\Gamma(\frac{4-p}{2})/48 > 0$. The last integral lies in $[0, \int \frac{u^6}{720}e^{-\mu u^2/2}u^{-p-1}du]$, so it is $O(\mu^{(p-6)/2})$. Hence $L(G_*) = \lambda\Psi(\lambda) = l_1\lambda^{(p-2)/2} - c_p'\lambda + O(\lambda^{(p-4)/2})$, and by Lemma 9.13

$$c_p'\big(\Phi(G_*) - m_p\lambda^{p/2}\big) = c_p'\lambda + L(G_*) + N(G_*) = (l_1 + n_1)\lambda^{(p-2)/2} + O(\lambda^{(p-4)/2}).$$

Finally $(l_1 + n_1)/c_p' = K_p$. Indeed, Legendre's duplication formula for $\Gamma(-p)$ and the reflection formula for $\Gamma(\frac{1-p}{2})\Gamma(\frac{1+p}{2})$ give $c_p'm_p = -2^{-p/2-1}\Gamma(-\frac{p}{2})$. Together with $\Gamma(2 - \frac{p}{2}) = \frac{p(p-2)}{4}\Gamma(-\frac{p}{2})$ and $\Gamma(3 - \frac{p}{2}) = -\frac{p-4}{2}\Gamma(2 - \frac{p}{2})$ this yields $l_1 = c_p'm_p\frac{p(p-2)}{24}$ and $n_1 = c_p'm_p\frac{p(p-2)(p-4)}{72}$, whose sum is $c_p'm_p\frac{p(p-1)(p-2)}{72}$. Scaling back ($c^p\lambda^{p/2} = B^{p/2}$, $c^p\lambda^{(p-2)/2} = B^{p/2}/\lambda$, $c^p\lambda^{(p-4)/2} = B^{p/2}\lambda^{-2}$) proves the theorem. □

Remark 9.14 (On the constants in Theorem J; numerical). Of the constant C_N, the part multiplying $\lambda^{(p-4)/2}$ is small; at $p = 4.5$ it is below 0.02. The bulk is the λ-independent term $3 \cdot 2^p/p \approx 15.1$ in the bound for N_3, from $t > \frac{1}{2}$, which was absorbed using $\lambda \ge 1$. Keeping it separate, the upper bound of Theorem J at $p = 4.5$ falls below the first correction $K_pB^{p/2}/\lambda$ once $\lambda \gtrsim 10^2$. The function φ_λ of Lemma 9.12 satisfies $\varphi_\lambda(s) = \lambda^{1/\beta}\varphi_1(s\lambda^{-1/\beta})$, so its concavity for all λ is equivalent to the case $\lambda = 1$.

Exactness in the Gaussian regime: proof of Theorem K. We keep $c = 1$ and the notation of the two previous proofs; in particular $\hat\omega$, C_ω and C_M are as in Lemmas 9.7 and 9.8, $Y = Y_G$, and $\Omega = \Omega(G) := \mathbb{E}\hat\omega(Y_G)$. Write $\delta\psi := \psi_G - \psi_{G_*} = \int e_t\,dH$ (Lemma 9.8), and note $\psi_r(G) - \psi_r(G_*) = \delta\psi$.

Lemma 9.15. *For all $t > 0$ and $y \in \mathbb{R}$,*

$$|e_t(y)| \le C_M \min(t^2, t^3)\,\hat\omega(y), \qquad \textit{hence} \qquad |\delta\psi(t)| \le C_M \min(t^2, t^3)\,\lambda\,\Omega.$$

Proof. For $t \ge 1$ this is Lemma 9.8. Let $t < 1$. On $[\frac{1}{2}, 2]$, $|e_t''| \le |h_t''| + 4|h_t'(1)| \le t^4/12 + 4t^3/6 \le \frac{3}{4}t^3$, so $|e_t(y)| \le \frac{3}{8}t^3(y-1)^2 \le \frac{3}{2}t^3\omega_0(y)$. Off $[\frac{1}{2}, 2]$, $|h_t(y) - h_t(1)| \le \frac{t^3}{6}|y-1| \le \frac{t^3}{3}(1 + |y|^{p-2})$ and $\frac{|h_t'(1)|}{p-2}\big||y|^{p-2} - 1\big| \le \frac{t^3}{12}(1 + |y|^{p-2})$. So $|e_t(y)| \le \frac{5}{12}t^3(1 + |y|^{p-2}) \le \frac{5}{12}\max(1, C_\omega)t^3\hat\omega(y)$. Both constants are below C_M. □

Lemma 9.16 (A one-sided bound for the imaginary part)**.** *Let $t_1 := \min\{0.45, (10/(3C_\omega))^{1/2}\}$ and $c_B := \min\{1/108, 1/(24\max(1, C_\omega))\}$. Then $\operatorname{Im} e_t(y) \ge c_Bt^3\hat\omega(y)$ for all $0 < t \le t_1$ and $y \in \mathbb{R}$.*

Proof. Let $S(u) := (u - \sin u)/u^2$ ($S(0) = 0$), an odd function, and $R(u) := u/6 - S(u)$. From $u - \sin u = u^3\int_0^1 \frac{(1-v)^2}{2}\cos(uv)\,dv$ we get $R(u) = u\int_0^1 \frac{(1-v)^2}{2}(1 - \cos uv)\,dv$. Hence, for $u \ge 0$, $0 \le R(u) \le \min(u^3/120, u/6)$, and, using $1 - \cos x \le x^2/2$, $|\sin x| \le |x|$ and $\int_0^1 \frac{(1-v)^2}{2}v^2dv = \frac{1}{60}$, also $|R'(u)| \le u^2/40$ and $|R''(u)| \le u/20$. Since $\operatorname{Im} h_t(y) = (\sin ty - ty)/y^2 = -t^2S(ty)$, we have

$$\operatorname{Im} e_t(y) = t^2F_t(y), \qquad F_t(y) := S(t) - S(ty) + tS'(t)\frac{|y|^{p-2} - 1}{p-2}.$$

For $y \ge 0$ (with $h_t(0) = -t^2/2$, so that $S(0) = R(0) = 0$), inserting $S = u/6 - R$ gives $F_t(y) = \frac{t}{6}\omega_0(y) + \Gamma_t(y)$ with $\Gamma_t(y) := R(ty) - R(t) - tR'(t)\frac{y^{p-2}-1}{p-2}$. On $[\frac{1}{2}, 2]$, $\Gamma_t(1) = \Gamma_t'(1) = 0$ and $|\Gamma_t''(y)| = |t^2R''(ty) - tR'(t)(p-3)y^{p-4}| \le \frac{t^3}{10} + \frac{t^3}{10}$, so $|\Gamma_t(y)| \le \frac{t^3}{10}(y-1)^2 \le \frac{2}{5}t^3\omega_0(y) \le \frac{t}{12}\omega_0(y)$, since $t \le 0.45$. Off $[\frac{1}{2}, 2]$, $R(ty) \ge 0$ gives $\Gamma_t(y) \ge -\frac{t^3}{120} - \frac{t^3}{80}(1 + y^{p-2}) \ge -\frac{t^3}{40}(1 + y^{p-2}) \ge -\frac{C_\omega t^3}{40}\omega_0(y) \ge$

$-\frac{t}{12}\omega_0(y)$, since $t^2 \le 10/(3C_\omega)$; at $y=0$ this uses that the supremum defining C_ω includes $s=0$. Thus $F_t(y) \ge \frac{t}{12}\omega_0(y)$ for $y \ge 0$. For $y<0$, oddness of S gives $F_t(y) = F_t(|y|) + 2S(t|y|) \ge \frac{t}{12}\omega_0(|y|) + 2S(t|y|)$. Here $S \ge 0$ on $[0,\infty)$, and $2S(u) \ge \frac{u}{3} - \frac{u^3}{60} \ge \frac{u}{6}$ for $0 \le u \le 3$. For $|y| \in [\frac12, 2]$ (so $t|y| \le 1$) this gives $F_t(y) \ge \frac{t}{12}\omega_0(|y|) + \frac{t}{12} \ge \frac{t}{108}\hat\omega(y)$, because $1 + |y|^{p-2} \le 9$. For $|y| \notin [\frac12, 2]$, $F_t(y) \ge \frac{t}{24}\omega_0(|y|) + \frac{t}{24C_\omega}(1+|y|^{p-2})$. In all cases $F_t(y) \ge c_B t\,\hat\omega(y)$. □

Proof of Theorem K. Let $\lambda \ge \max(1, t_1^{-2})$ and $G \in \mathscr{G}_\lambda$. All constants $C_1, C_2, \dots$ below depend only on p. By Lemmas 9.11 and 9.12, and with $N = N_s - N_c - N_3$ as in Lemma 9.13,

$$c_p'\big(\Phi(G) - \Phi(G_*)\big) \le \big(N_s(G) - N_s(G_*)\big) + |N_c(G) - N_c(G_*)| + |N_3(G) - N_3(G_*)|. \tag{12}$$

Step 1: variation of N_c and N_3. By (11) and Lemma 9.15, $|(\operatorname{Re}\psi_r(G))^2 - (\operatorname{Re}\psi_r(G_*))^2| \le |\delta\psi|\frac{\lambda t^4}{12} \le \frac{C_M}{12}\lambda^2\Omega\, t^7$, so $|N_c(G) - N_c(G_*)| \le \frac{C_M}{24}\lambda^2\Omega\int_0^\infty e^{-g}t^{6-p}dt = C_1\lambda^{(p-3)/2}\Omega$. For N_3 and $t \le \frac12$, every point z of the segment $[\psi_r(G_*), \psi_r(G)]$ has $|z| \le \lambda t^3/6$ and $\operatorname{Re} z \le g - \frac{11}{16}g$, by the bound $a \ge \frac{11}{16}g$ applied to both measures. Since $|\rho'(z)| = |e^z - 1 - z| \le \frac{|z|^2}{2}e^{\max(\operatorname{Re} z, 0)}$, we get $e^{-g}|\rho(\psi_r(G)) - \rho(\psi_r(G_*))| \le |\delta\psi| e^{-11g/16}\frac{\lambda^2 t^6}{72} \le \frac{C_M}{72}\lambda^3\Omega\, t^9 e^{-11g/16}$. For $t > \frac12$ we write $e^{-g}\rho(\psi_r) = e^{\psi_G} - e^{-g}(1 + \psi_r + \psi_r^2/2)$. For $\operatorname{Re} z_0, \operatorname{Re} z_1 \le 0$,

$$|e^{z_1} - e^{z_0}| \le |z_1 - z_0| \int_0^1 e^{(1-s)\operatorname{Re} z_0} ds \le |z_1 - z_0| \min\big(1, 1/|\operatorname{Re} z_0|\big).$$

We apply this with $z_0 = \psi_{G_*}$, $|\operatorname{Re} z_0| = a_*(t) = \lambda(1-\cos t)$, and use $|\psi_r| \le 2g$ for the remaining terms. This gives $e^{-g}|\rho(\psi_r(G)) - \rho(\psi_r(G_*))| \le |\delta\psi|\big(\min(1, 1/a_*(t)) + e^{-g}(1+2g)\big)$. Since $1 - \cos t \ge \frac{2}{\pi^2}d(t)^2$ with $d(t) = \operatorname{dist}(t, 2\pi\mathbb{Z})$, we have $\int_{\mathbb{R}} \min(1, \frac{\pi^2}{2\lambda u^2})du = 2\sqrt{2}\,\pi\lambda^{-1/2}$. Summing over the neighbourhoods $[(2k-1)\pi, (2k+1)\pi]$ of $2\pi k$ gives

$$J(\lambda) := \int_{1/2}^\infty t^{1-p} \min\Big(1, \frac{1}{a_*(t)}\Big)dt \le 2\sqrt{2}\,\pi^{2-p}\zeta(p-1)\,\lambda^{-1/2} + \frac{\pi^2 2^{p-1}}{p}\,\lambda^{-1}.$$

On $[\frac12, \pi]$ we used $d(t) = t$. Moreover $e^{-g}(1+2g) \le 3e^{-\lambda t^2/4}$. With $|\delta\psi| \le C_M t^2\lambda\Omega$, the part $t > \frac12$ is at most $C_M\lambda\Omega\,(J(\lambda) + 3\int_{1/2}^\infty e^{-\lambda t^2/4}t^{1-p}dt) = O\big((\lambda^{1/2}+1)\Omega\big)$. Integrating, and using $\lambda^{1/2} \le \lambda^{(p-3)/2}$ for $\lambda \ge 1$, we get $|N_3(G) - N_3(G_*)| \le C_2\lambda^{(p-3)/2}\Omega$.

Step 2: the case $\Omega(G) \le \Omega_1 := 1/(7C_M)$. Put $\sigma(t) := t - \sin t$, so $\operatorname{Im}\psi_{G_*} = -\lambda\sigma$, and put $\varepsilon_t := \mathbb{E}\operatorname{Im} e_t(Y)$, so $\operatorname{Im}\psi_G = -\lambda\sigma + \lambda\varepsilon_t$. Then $(\operatorname{Im}\psi_{G_*})^2 - (\operatorname{Im}\psi_G)^2 = \lambda^2\varepsilon_t(2\sigma - \varepsilon_t)$. For $t \le t_1$, Lemmas 9.16 and 9.15 give $c_B t^3\Omega \le \varepsilon_t \le C_M t^3\Omega$. Moreover $\sigma \ge t^3/6 - t^5/120 \ge t^3/7$, so $2\sigma - \varepsilon_t \ge t^3/7$ and $(\operatorname{Im}\psi_{G_*})^2 - (\operatorname{Im}\psi_G)^2 \ge \frac{c_B}{7}\lambda^2\Omega\, t^6$. For $t > t_1$, $|\lambda^2\varepsilon_t(2\sigma - \varepsilon_t)| \le \lambda^2 C_M t^3\Omega(\frac{t^3}{3} + C_M t^3\Omega_1) \le \frac{C_M}{2}\lambda^2\Omega t^6$. Hence, using $\int_0^{t_1} e^{-\lambda t^2/2}t^{5-p}dt \ge \lambda^{-(6-p)/2}e^{-1/2}/(6-p)$ (valid as $t_1\sqrt{\lambda} \ge 1$) and $e^{-\lambda t^2/2} \le e^{-\lambda t_1^2/4}e^{-t^2/4}$ for $t \ge t_1$, $\lambda \ge 1$,

$$N_s(G_*) - N_s(G) \ge c_s\lambda^{(p-2)/2}\Omega - C_3\lambda^2 e^{-\lambda t_1^2/4}\Omega, \qquad c_s := \frac{c_B e^{-1/2}}{14(6-p)}.$$

With (12) and Step 1, $c_p'(\Phi(G) - \Phi(G_*)) \le \lambda^{(p-2)/2}\Omega\big(-c_s + C_3\lambda^{(6-p)/2}e^{-\lambda t_1^2/4} + (C_1+C_2)\lambda^{-1/2}\big)$. Fix $\lambda_{\mathrm{near}}(p) \ge \max(1, t_1^{-2})$ such that the bracket is negative for all $\lambda \ge \lambda_{\mathrm{near}}(p)$; such a number exists because the bracket tends to $-c_s$. For $\lambda \ge \lambda_{\mathrm{near}}(p)$ we get $\Phi(G) \le \Phi(G_*)$, with equality only if $\Omega = 0$, i.e. $G = G_*$.

Step 3: the remaining case. Let $\check G := G(-\,\cdot\,)$; then $\Phi(\check G) = \Phi(G)$ and $Y_{\check G} = -Y_G$. If $\Omega(G) \le \Omega_1$ or $\Omega(\check G) \le \Omega_1$, Step 2 applies to G or to $\check G$. Otherwise choose $G' \in \{G, \check G\}$ with $\pi_+ \ge \pi_-$, where $\pi_\pm := \mathbb{P}(\pm Y' \in [\frac12, 2])$ and $Y' = Y_{G'}$. Then $\Omega(G') > \Omega_1$. Since $\mathbb{E}Y'_\pm \ge \pi_\pm/2$,

$$|\mathbb{E}Y'| = \mathbb{E}|Y'| - 2\min(\mathbb{E}Y'_+, \mathbb{E}Y'_-) \le 1 - \mathbb{E}\omega_0(|Y'|) - \pi_-,$$

and, as in the proof of Lemma 9.7, $\Omega(G') \le (1+C_\omega)\mathbb{E}\omega_0(|Y'|) + 9\pi_-$. Hence $1-(\mathbb{E}Y')^2 \ge 1-|\mathbb{E}Y'| \ge \Omega(G')/(10+C_\omega) > \Omega_1/(10+C_\omega)$. Next, with $\sigma_t(y) := (ty-\sin ty)/y^2$ we have $\operatorname{Im}\psi_{G'}(t) = -\lambda\mathbb{E}\sigma_t(Y')$ and $\sigma_t(y) - t^3y/6 = -t^2R(ty)$. Interpolating $R(u) \le \min(u^3/120, u/6)$ ($u \ge 0$; R is odd, so $|R(u)| = R(|u|)$) with exponent $\theta = (p-3)/2$ gives $t^2|R(ty)| \le C_4 t^p|y|^{p-2}$, so $|\mathbb{E}\sigma_t(Y')| \le \frac{t^3}{6}|\mathbb{E}Y'| + C_4t^p$. Squaring and integrating, $N_s(G') \le (\mathbb{E}Y')^2N_s^+ + C_5\lambda^{1/2}$. Lemma 9.13 gives $N_s(G_*) \ge N_s^+ - C_N\lambda^{(p-4)/2}$, together with the bounds $0 \le N_c(\cdot) \le C_N\lambda^{(p-4)/2}$ and $|N_3(\cdot)| \le C_N\lambda^{(p-4)/2}$, valid for every measure in $\mathscr{G}_\lambda$. So (12) gives

$$c_p'\big(\Phi(G') - \Phi(G_*)\big) \le -\frac{\Omega_1 n_1}{10+C_\omega}\lambda^{(p-2)/2} + C_5\lambda^{1/2} + 5C_N\lambda^{(p-4)/2},$$

which is negative for all $\lambda \ge \lambda_{\mathrm{far}}(p)$, for a suitable $\lambda_{\mathrm{far}}(p) < \infty$. Hence $\Phi(G) = \Phi(G') < \Phi(G_*)$.

With $\lambda_1(p) := \max\{1, t_1^{-2}, \lambda_{\mathrm{near}}(p), \lambda_{\mathrm{far}}(p)\}$ we have shown that $\Phi(G) \le \Phi(G_*)$ for all $G \in \mathscr{G}_\lambda$, with equality only for $G \in \{\lambda\delta_1, \lambda\delta_{-1}\}$. Poissonization and [7, Lemma 2.1] conclude the proof, as for Theorem I. □

Remark 9.17 (Size of λ_1; numerical). We evaluated the explicit constants of the proof with the script `numerics/lambda1_constants.py`. At $p = 4.05, 4.2, 4.5, 4.8$ they give $\lambda_{\mathrm{near}} \approx 10^{13.0}, 10^{12.3}, 10^{11.8}, 10^{11.4}$ and $\lambda_{\mathrm{far}} \approx 10^{9.3}, 10^{8.1}, 10^{6.8}, 10^{6.0}$. So $\lambda_1(p) \approx 10^{11}$–$10^{13}$, and it stays bounded as $p \downarrow 4$, unlike $\lambda_0(p)$. The refined Lipschitz bound in Step 1 matters here. With $|e^{z_1} - e^{z_0}| \le |z_1 - z_0|$ alone, a term of order $\lambda\Omega$ would remain; that gives $\lambda_1 \approx 10^{29}$ at $p = 4.5$, with $\lambda_1 \to \infty$ as $p \downarrow 4$. The remaining loss comes from the constants C_M and c_B. Numerically, $\operatorname{Im} e_t(y)/(t^3\hat{\omega}(y))$ stays above 0.04 for $t \le t_1$ (`numerics/check_K1.py`), whereas $c_B \approx 0.005$.

Remark 9.18 (A conditional reduction for all λ; numerical). By Lemmas 9.11 and 9.12, Pinelis' conjecture for a given λ follows from the inequality $N(G) \le N(G_*)$ for all centred $G \in \mathscr{G}_\lambda$; Poissonization only produces centred measures. We tested it at $p = 4.5$ on two families. The first is 150 random two- and three-atom G for each $\lambda \in \{0.05, 0.3, 1, 3, 10\}$. The second is a grid of 343 two-atom G for each λ, of which 7 coincide with $\lambda\delta_{-1}$. For $\lambda \ge 0.3$ the inequality held in every non-degenerate case. At $\lambda = 0.05$ it failed for 64 of the 150 random and 184 of the 343 grid measures, although $\Phi(G) < \Phi(G_*)$ held for all non-degenerate measures of both families. So it is not a substitute for Theorem I when λ is small. The scripts are `numerics/check_N_sign.py` (outputs `nsign_*.txt`) and `numerics/check_large_lambda.py`. The latter also checks Lemma 9.11 against exact Poisson summation, to ten digits, and the concavity of φ_1 on a logarithmic grid.

10. Discussion; what remains open

(1) **Where $p \ge 3$ is used.** Only in Corollary 2.5, through convexity of $|v+x|^{p-2}$, which is needed to apply Lemma 2.4. The bound (4) for compound Poisson laws holds for all $2 < p < 4$.

(2) **Why $p = 4$ is the threshold.** For $2 < p < 4$ the representing kernel $\cos s - 1 + s^2/2$ is nonnegative and the comparison $F(a) \le F(g)$ goes the right way. For $4 < p < 6$ the kernel changes to $-(\cos s - 1 + s^2/2 - s^4/24)$ and the same comparison reverses (Proposition G).

(3) **Open: $4 < p < 5$ with $B^{p/2}/A$ not small.** Pinelis' conjecture $\mathcal{E}_{p;A,B} = c^p\mathbb{E}|\Pi_\lambda - \lambda|^p$ is proved here only for $\lambda \le \lambda_0(p)$ (Theorem I), with a tiny $\lambda_0(p)$, and for $\lambda \ge \lambda_1(p)$ (Theorem K), with $\lambda_1(p) \approx 10^{12}$. It remains open for $\lambda \in (\lambda_0(p), \lambda_1(p))$. For $1 \le \lambda < \lambda_1(p)$, Theorem J confirms it up to $O(B^{p/2}\lambda^{-2})$, and Remark 9.18 gives a sufficient condition, $N(G) \le N(G_*)$, that held in all our tests with $\lambda \ge 0.3$. Making λ_0, λ_1 reasonable would need sharper constants in Lemmas 9.7, 9.8 and 9.16; closing the gap requires a non-perturbative idea. In particular the companion hypothesis of [6] for $t \ge 4$, $C(t) = \mathbb{E}|\Pi_1 - 1|^t$, remains open for $4 < t < 5$. Under Pinelis' conjecture it is the case $\lambda = 1$: numerically, $\mathbb{E}|\Pi_\lambda - \lambda|^t/\max(\lambda, \lambda^{t/2})$ is maximal at $\lambda = 1$. That value lies inside the gap $(\lambda_0(p), \lambda_1(p))$. The other partial results are Theorem H (the conjecture holds up to an additive $C_pB^{p/2}$,

which is of lower order than both sides when $B^{p/2}/A \to 0$), Proposition 9.6 (second-order optimality of the Poisson law along every ray in the rare-jump regime), and Proposition 9.3 (a reduction to a "skewness inequality" bounding $\mathbb{E}|\tilde{X}_G|^p - \mathbb{E}|X_{G_s}|^p$ by its Poisson value). After Poissonization, $c_p'(\mathbb{E}|X_G|^p - A) = \int_0^\infty (1 - a - e^{-a}\cos b + g^2/2) t^{-p-1} dt$. The Poisson law is not a pointwise maximizer of this integrand, so a variational argument is needed. Numerically (grid scans of two-atom Lévy measures at $p = 4.5$ and $\lambda_{\text{cand}} \in \{0.05, 0.3, 1, 3, 10, 40\}$; random and differential-evolution searches over up to three atoms plus a Gaussian part at $p \in \{4.1, \dots, 4.9\}$) the Poisson value was never beaten; relative margins of the best two-atom competitors ranged from 10^{-9} (nearly merged same-side atoms) to $5 \cdot 10^{-3}$ (two-sided).

(4) **Role of the moment-curve viewpoint (heuristic).** The first-order (KKT) conditions identify the support of an optimal Lévy measure with the contact set of the planar curve $v \mapsto (v, \Psi_W(\pm v^{1/(p-2)}))$ with its upper hull, in the sense of [3]. Here $\Psi_W(u) = \mathbb{E}[f(W+u) - f(W) - uf'(W)]/u^2$. Convexity of this curve on the skew side and its failure on the anti-skew side (observed numerically, not proved) is consistent with the failure of the extension with an extra skewed summand (Theorem E), whose proof is a direct Taylor argument. The final proof of Theorem A bypasses the curve analysis entirely.

(5) **Open: the extra-summand version for $3 < p < 4$.** Theorems D and E leave open: symmetric X whose characteristic function changes sign (e.g. Rademacher X); non-symmetric X with $\mathbb{E}[|X|^{p-3} \operatorname{sign} X] = 0$; and X that is unbounded or not bounded away from 0 (Theorem E(b) needs both). For $X = \pm x_0$ Rademacher we found no violation in about 900 random compound Poisson configurations at $p \in \{3.2, 3.5, 3.8\}$ (the gap approached 0 only in degenerate limits), which suggests that the extension holds for all symmetric X.

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